

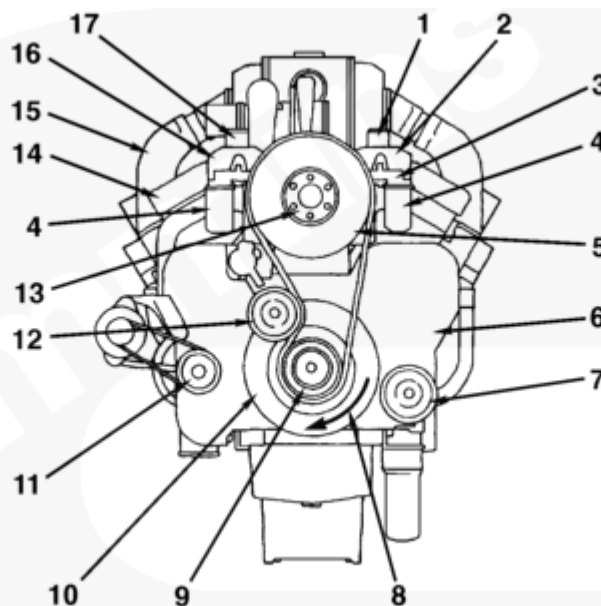
Trainings ▾

Engine Views

Industrial Applications

The illustrations show the locations of the major external engine components, and filters, along with service and maintenance points. Some external components will be at different locations for different engine models.

©Cummins Inc



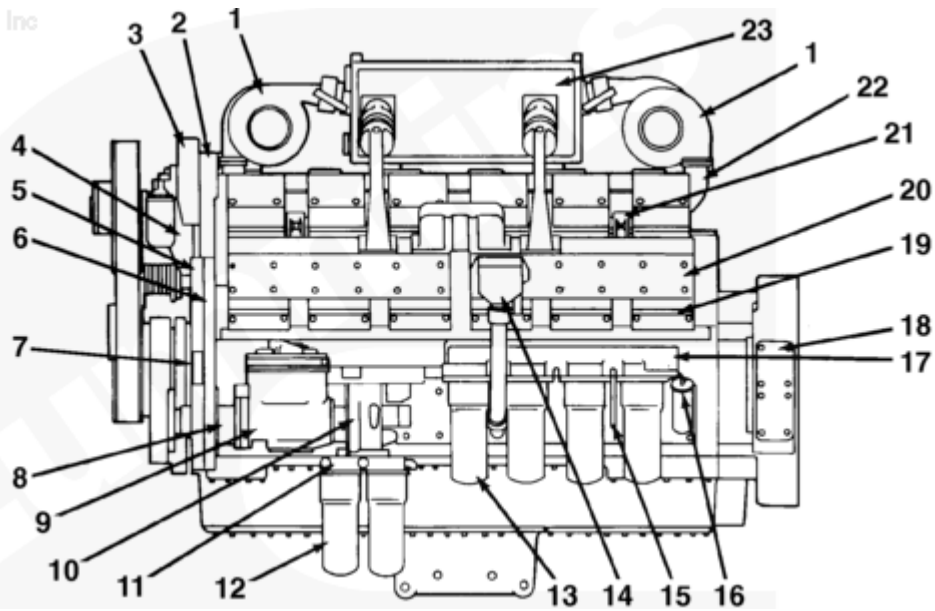
00600374

Front View - KT38

1. Left-bank water outlet
2. Left-bank thermostat housing
3. Coolant filter head
4. Coolant filters
5. Fan pulley
6. Front gear cover
7. Accessory drive pulley
8. Direction of rotation
9. Crankshaft pulley
10. Vibration damper
11. Alternator drive pulley
12. Fan idler pulley
13. Fan hub
14. Rocker lever cover
15. Air crossover

16. Right-bank thermostat housing
17. Right-bank water outlet.

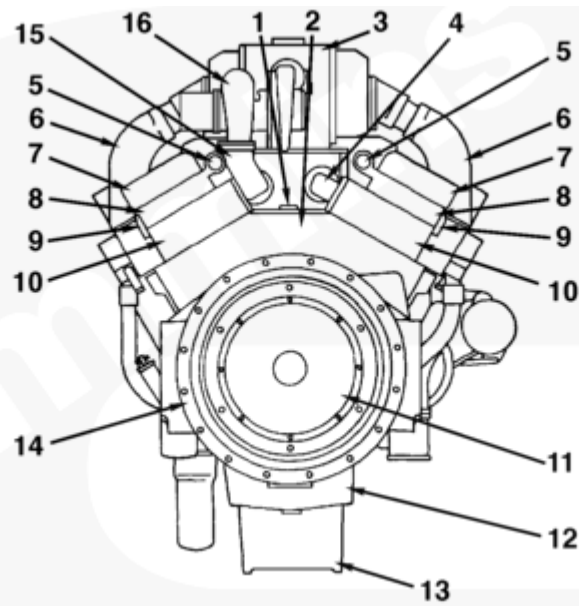
©Cummins Inc



ew600gb

Left View - KT38

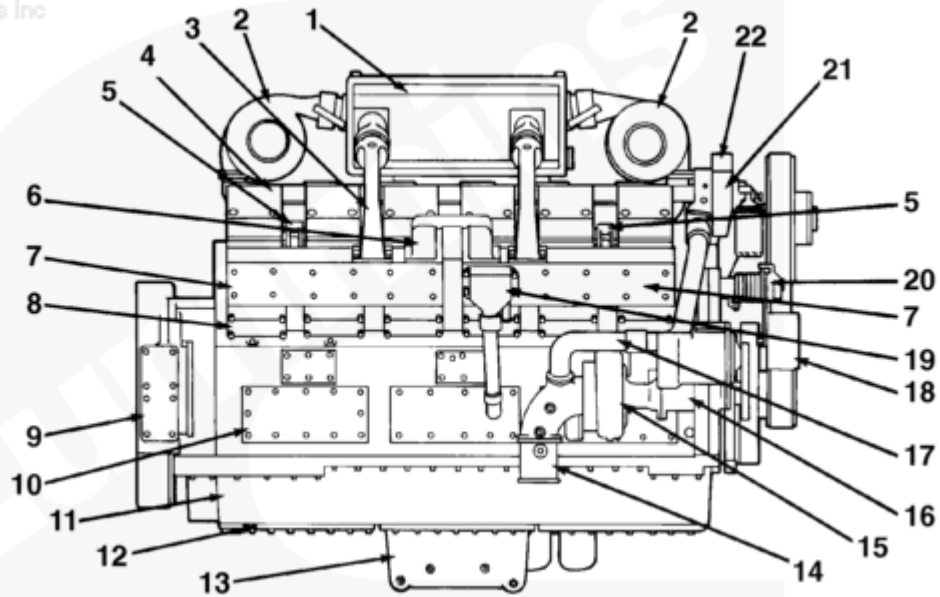
1. Turbocharger
2. Thermostat housing support
3. Thermostat housing
4. Fan hub support
5. Front gear cover
6. Front gear housing
7. Dataplate
8. Air compressor drive
9. Air compressor
10. Fuel pump
11. Fuel filter head
12. Fuel filter
13. Full-flow lubricating oil filter
14. Crankcase breather
15. Oil level gauge
16. Oil fill port
17. Lubricating oil filter head
18. Flywheel housing
19. Cam follower cover
20. Intake manifold
21. Lifting bracket
22. Exhaust manifold
23. Air transfer housing.



00600373

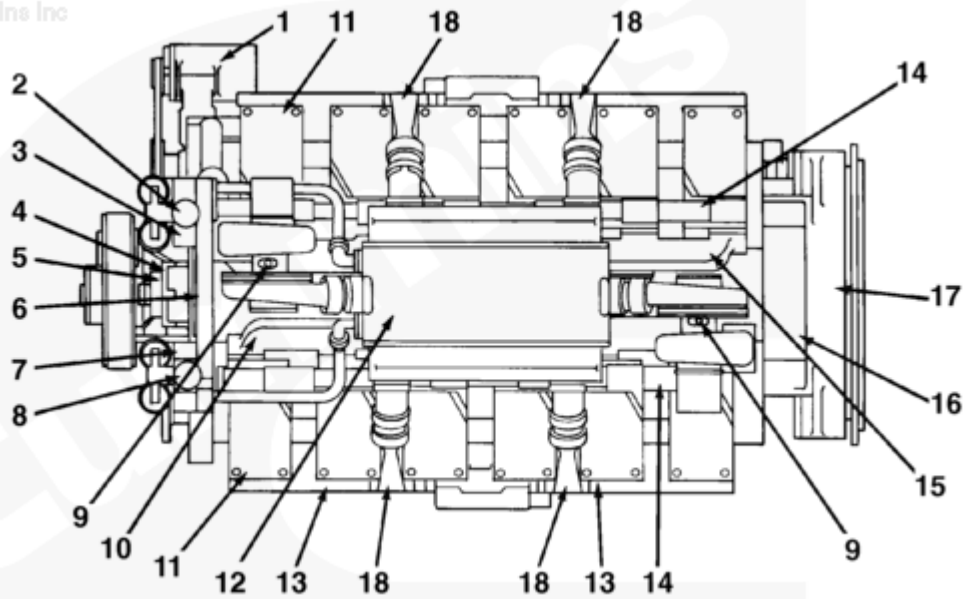
Rear View - KT38

1. Oil cooler cover
2. Cylinder block
3. Air transfer housing
4. Right-bank exhaust manifold
5. Water outlet passage
6. Air crossover
7. Rocker lever cover
8. Rocker lever housing
9. Fuel manifold
10. Cylinder head
11. Flywheel
12. Oil pan adapter
13. Oil pan
14. Flywheel housing
15. Left-bank exhaust manifold
16. Turbocharger.



Right View - KT38

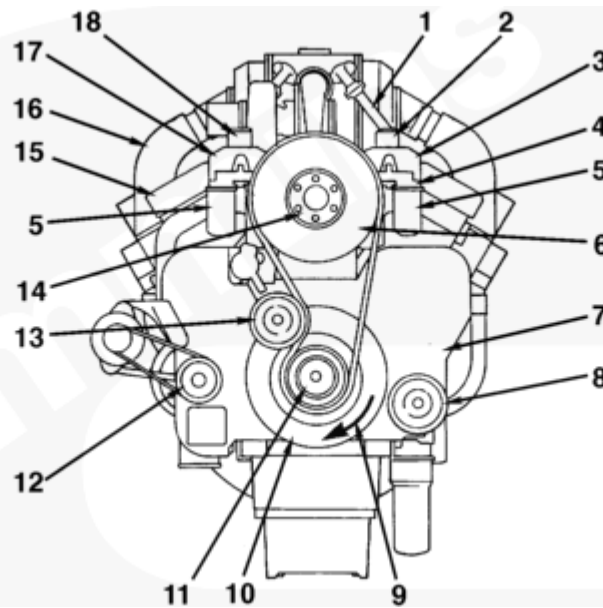
1. Air transfer housing
2. Turbocharger
3. Air crossover connection
4. Rocker lever cover
5. Lifting bracket
6. Air intake connection
7. Intake manifold
8. Cam follower cover
9. Flywheel housing
10. Hand hole cover
11. Oil pan adapter
12. Plate, oil pan adapter
13. Oil pan
14. Water inlet connection
15. Water pump
16. Water pump drive
17. Water bypass tube
18. Fan belt idler pulley
19. Crankcase breather
20. Fan belt idler assembly
21. Thermostat housing
22. Water outlet connection.



ew600gc

Top View - KT38

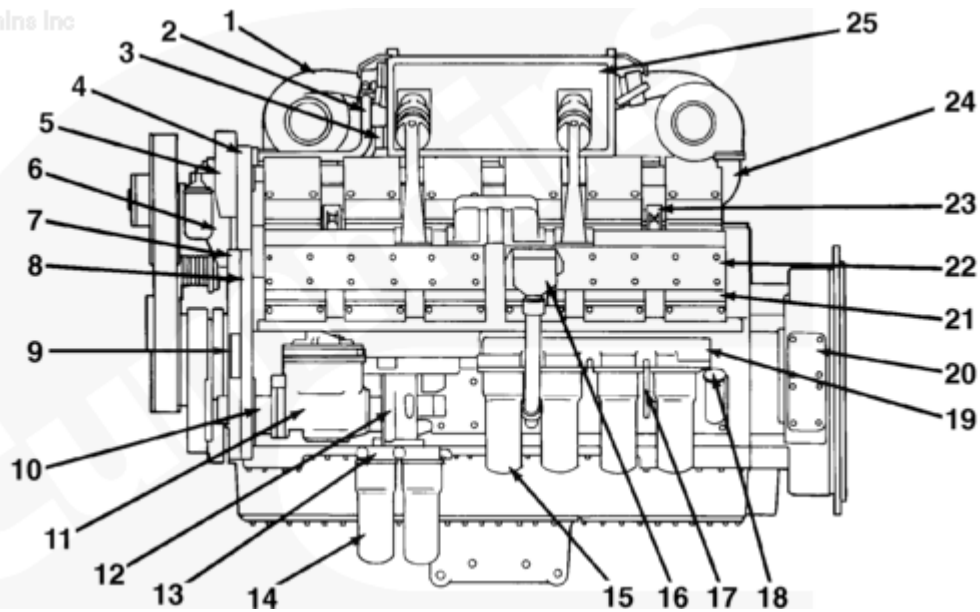
1. Alternator
2. Right-bank water outlet
3. Right-bank thermostat housing
4. Fan hub bracket
5. Fan hub shaft
6. Thermostat housing support
7. Left-bank thermostat housing
8. Left-bank water outlet
9. Turbocharger
10. Left-bank exhaust manifold
11. Rocker lever cover
12. Air transfer housing
13. Intake manifold
14. Water transfer tube
15. Right-bank exhaust manifold
16. Flywheel housing
17. Rocker lever housing
18. Air crossover.



00600376

Front View - KTA38 with Center Mount Aftercooler

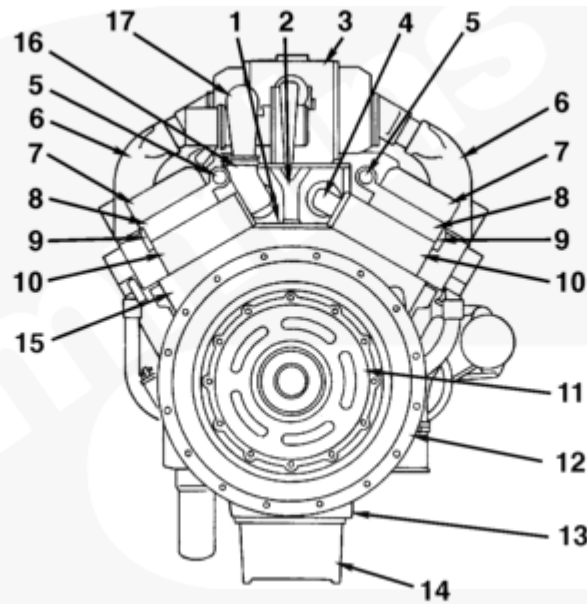
1. Aftercooler water outlet
2. Left-bank water outlet
3. Left-bank thermostat housing
4. Coolant filter head
5. Coolant filters
6. Fan pulley
7. Front gear cover
8. Accessory drive pulley
9. Direction of rotation
10. Vibration damper
11. Crankshaft pulley
12. Alternator drive pulley
13. Fan idler pulley
14. Fan hub
15. Rocker lever cover
16. Air crossover
17. Right-bank thermostat housing
18. Right-bank water outlet.



ew600gf

Left View - KTA38 with Center Mount Aftercooler

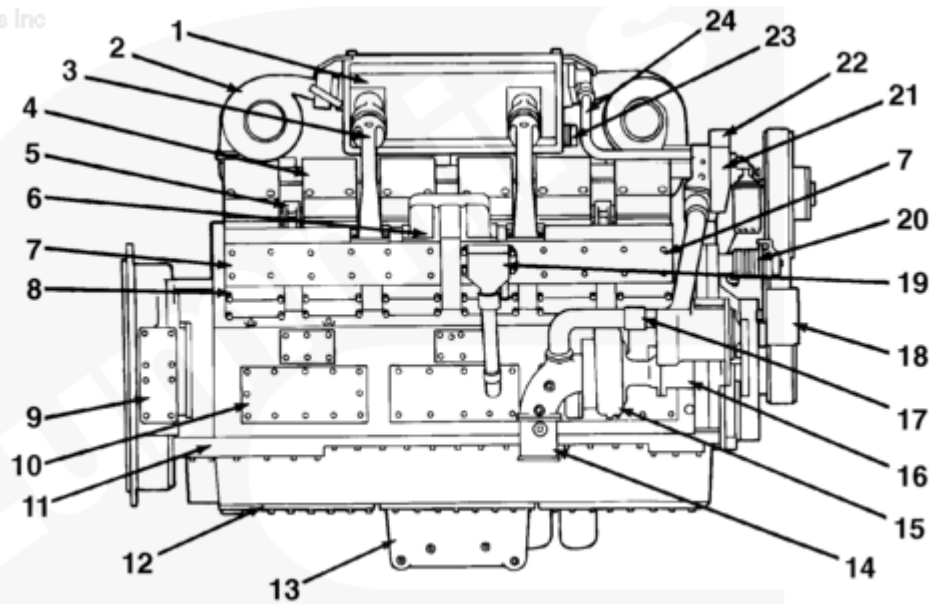
1. Turbocharger
2. Aftercooler water outlet
3. Aftercooler water inlet
4. Thermostat housing support
5. Thermostat housing
6. Fan hub support
7. Front gear cover
8. Gear housing
9. Dataplate
10. Air compressor drive
11. Air compressor
12. Fuel pump
13. Fuel filter head
14. Fuel filter
15. Full-flow lubricating oil filter
16. Crankcase breather
17. Oil level gauge
18. Oil fill port
19. Lubricating oil filter head
20. Flywheel housing
21. Cam follower cover
22. Intake manifold
23. Lifting bracket
24. Exhaust manifold
25. Aftercooler assembly.



00600375

Rear View - KTA38 with Center Mount Aftercooler

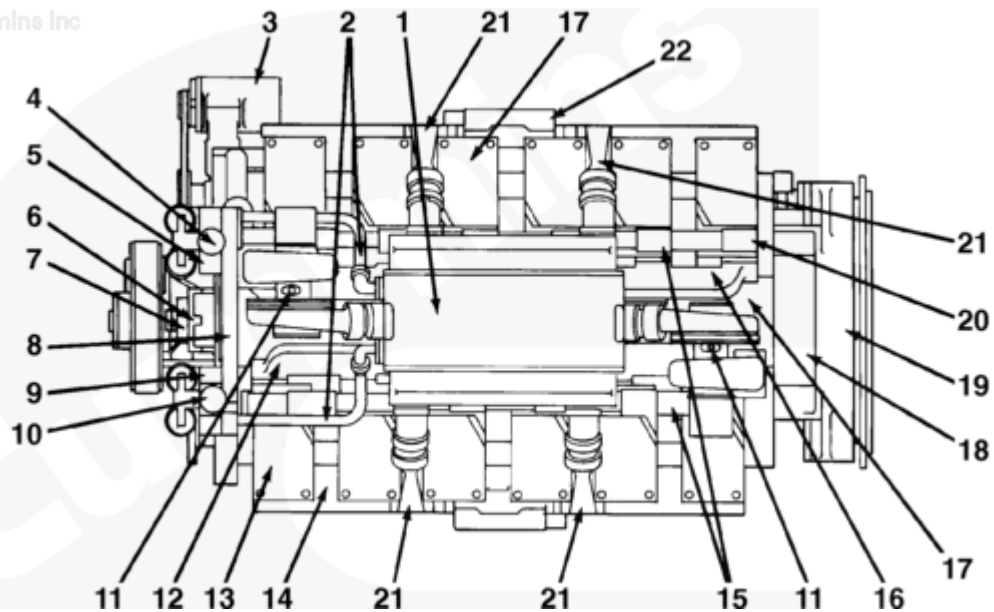
1. Oil Cooler Plate
2. Aftercooler water inlet tube
3. Aftercooler assembly
4. Right-bank exhaust manifold
5. Water outlet passage
6. Air crossover
7. Rocker lever cover
8. Rocker lever housing
9. Fuel manifold
10. Cylinder head
11. Flexplate
12. Flywheel housing
13. Oil pan adapter
14. Oil pan
15. Cylinder block
16. Left-bank exhaust manifold
17. Turbocharger.



ew600ge

Right View - KTA38 with Center Mount Aftercooler

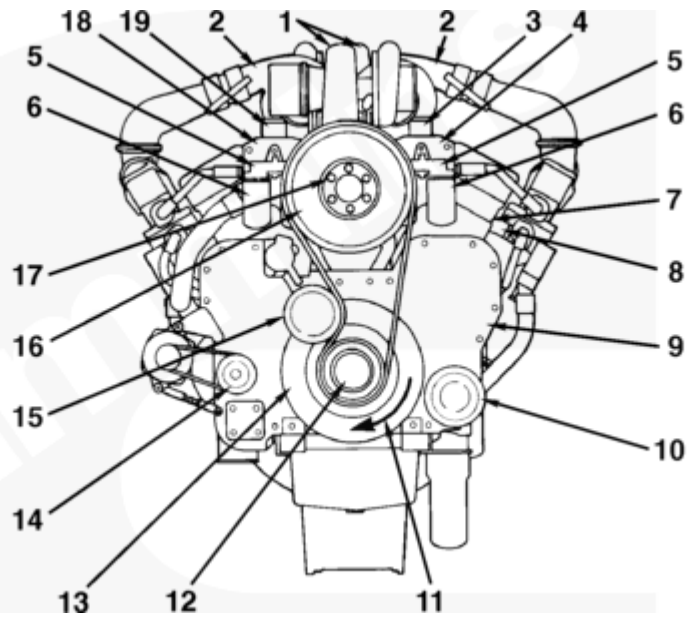
1. Aftercooler assembly
2. Turbocharger
3. Air crossover connection
4. Rocker lever cover
5. Lifting bracket
6. Air intake connection
7. Intake manifold
8. Cam follower cover
9. Flywheel housing
10. Hand hole cover
11. Oil pan adapter
12. Oil pan adapter cover
13. Oil pan
14. Water inlet connection
15. Water pump
16. Water pump drive
17. Water bypass tube
18. Fan belt idler pulley
19. Crankcase breather
20. Fan belt idler assembly
21. Thermostat housing
22. Water outlet connection
23. Aftercooler water inlet tube
24. Aftercooler water outlet tube.



ew600gg

Top View - KTA38 with Center Mount Aftercooler

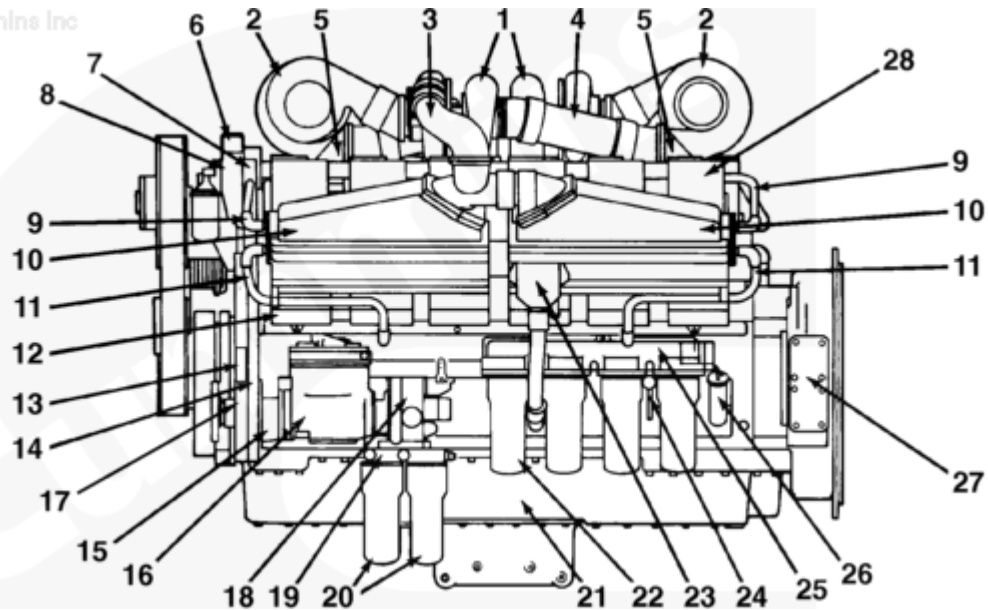
1. Aftercooler assembly
2. Aftercooler water outlet tube
3. Alternator
4. Right-bank water outlet
5. Right-bank thermostat housing
6. Fan hub support
7. Fan hub shaft
8. Thermostat housing support
9. Left-bank thermostat housing
10. Left-bank water outlet
11. Turbocharger
12. Left-bank exhaust manifold
13. Rocker lever cover
14. Intake manifold
15. Water tube
16. Right-bank exhaust manifold
17. Oil cooler plate
18. Rear gear housing
19. Flywheel housing
20. Rocker lever housing
21. Air crossover
22. Air intake connection.



00600378

Front View - KTTA38

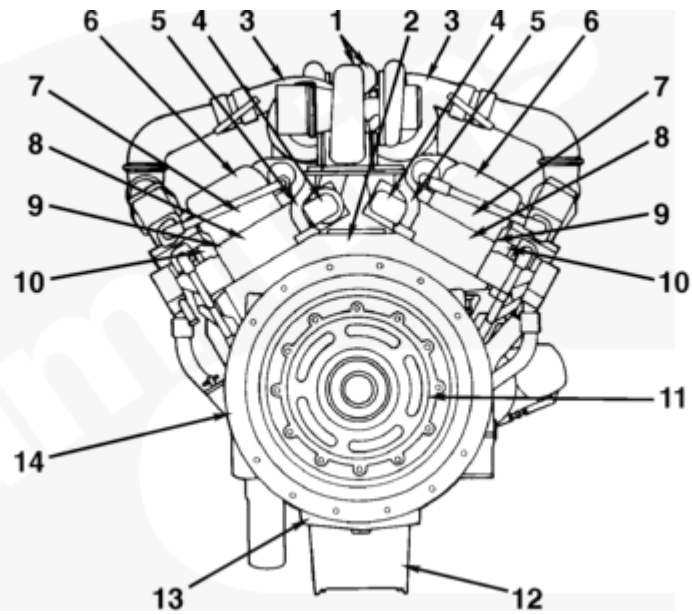
1. Low stage turbocharger
2. High stage turbocharger
3. Left-bank water outlet
4. Left-bank thermostat housing
5. Coolant filter head
6. Coolant filter
7. STC oil manifold
8. Fuel manifold
9. Front gear cover
10. Accessory drive pulley
11. Direction of rotation
12. Crankshaft pulley
13. Vibration damper
14. Alternator drive pulley
15. Fan idler pulley
16. Fan pulley
17. Fan hub
18. Right-bank thermostat housing
19. Right-bank water outlet.



ew600gj

Left View - KTTA38

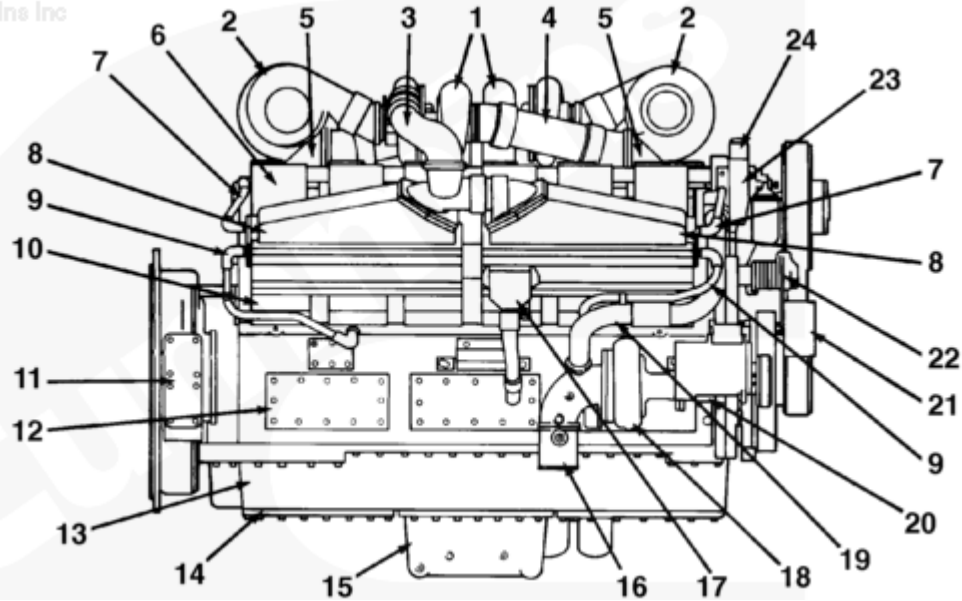
1. High stage turbocharger
2. Low stage turbocharger
3. Air crossover connection
4. Exhaust outlet pipe
5. Turbocharger support
6. Left-bank water outlet
7. Thermostat housing support
8. Left-bank thermostat housing
9. Aftercooler water outlet tube
10. Aftercooler assembly
11. Aftercooler water inlet tube
12. Cam follower cover
13. Dataplate
14. Front gear cover
15. Air compressor drive
16. Air compressor
17. Front cover
18. Fuel pump
19. Fuel filter head
20. Fuel filter
21. Oil pan adapter
22. Full-flow lubricating oil filter
23. Crankcase breather
24. Oil level gauge
25. Lubricating oil filter head
26. Oil fill port
27. Flywheel housing
28. Rocker lever cover.



00600377

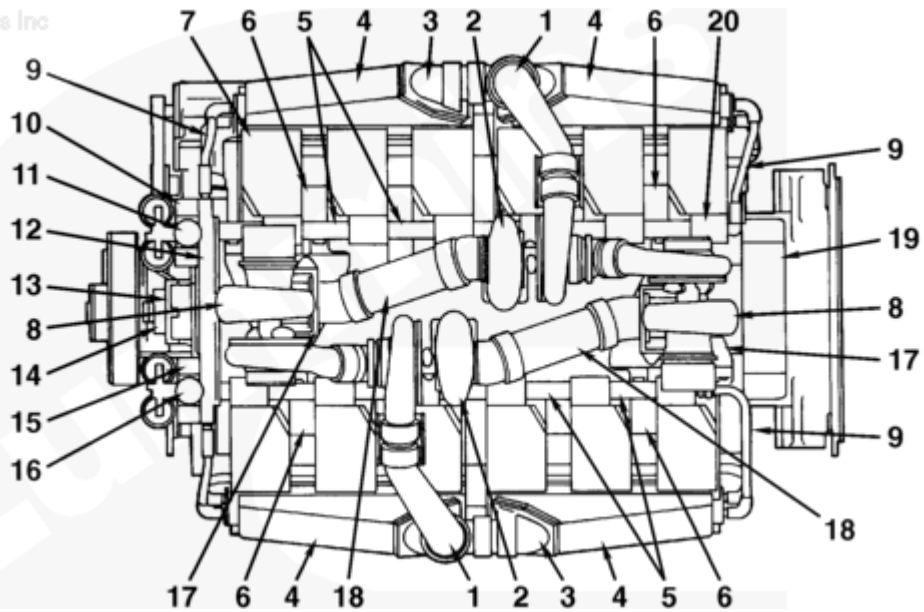
Rear View - KTTA38

1. Low stage turbocharger
2. Cylinder block
3. High stage turbocharger
4. Exhaust manifold
5. Turbocharger support
6. Rocker lever housing cover
7. Rocker lever housing
8. Cylinder head
9. STC oil manifold
10. Fuel manifold
11. Flexplate
12. Oil pan
13. Oil pan adapter
14. Flywheel housing.



Right View - KTTA38

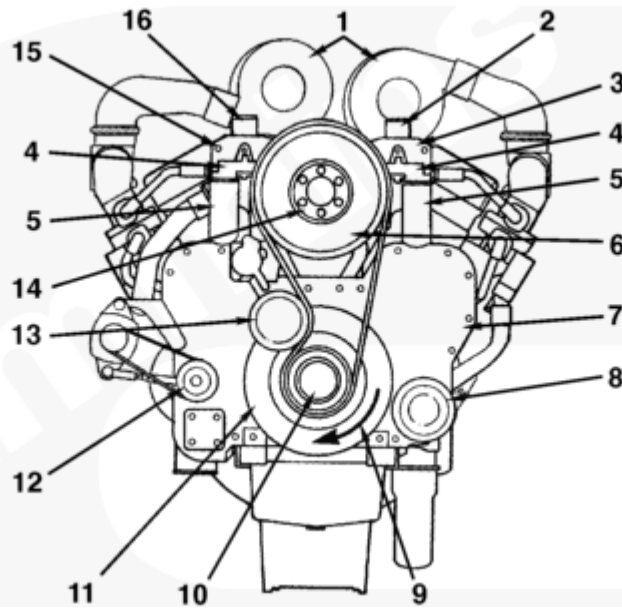
1. High stage turbocharger
2. Low stage turbocharger
3. Air crossover connection
4. Exhaust outlet pipe
5. Turbocharger support
6. Rocker lever cover
7. Aftercooler water outlet tube
8. Aftercooler assembly
9. Aftercooler water inlet tube
10. Cam follower cover
11. Flywheel housing
12. Hand hole cover
13. Oil pan adapter
14. Oil pan cover plate
15. Oil pan
16. Water inlet connection
17. Crankcase breather
18. Water pump
19. Water bypass tube
20. Water pump drive
21. Fan belt idler pulley
22. Fan belt idler arm assembly
23. Right-bank thermostat housing
24. Right-bank water outlet.



ew600gk

Top View - KTTA38

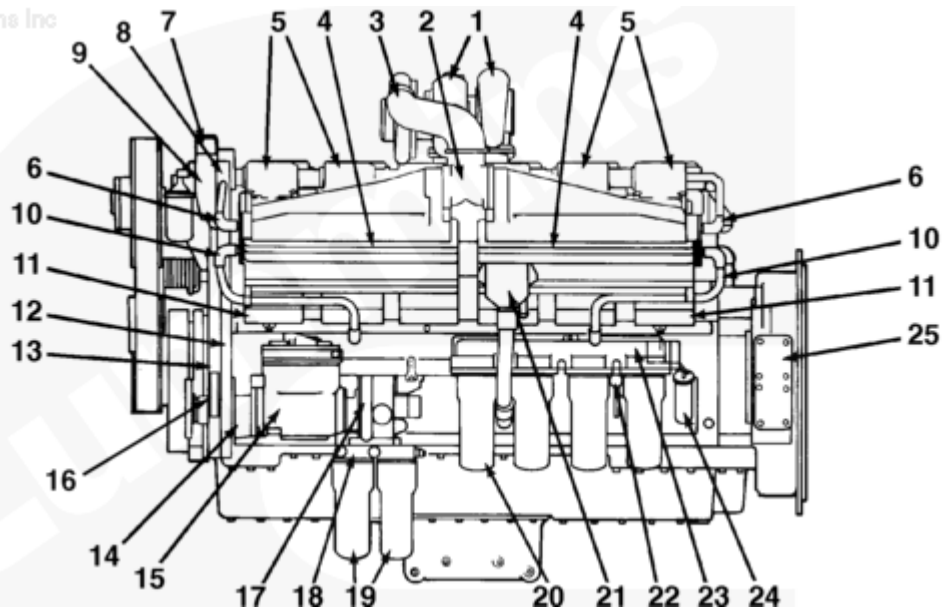
1. Air crossover connection
2. High stage turbocharger
3. Air connection
4. Aftercooler assembly
5. Water transfer tube
6. Lifting bracket
7. Rocker lever cover
8. Low stage turbocharger
9. Aftercooler water outlet tube
10. Right-bank thermostat housing
11. Right-bank water outlet
12. Thermostat housing support
13. Fan hub support
14. Fan hub shaft
15. Left-bank thermostat housing
16. Left-bank water outlet
17. Turbocharger support
18. Exhaust outlet pipe
19. Flywheel housing
20. Rocker lever housing.



00600380

Front View - KTA38 with Outboard Aftercoolers

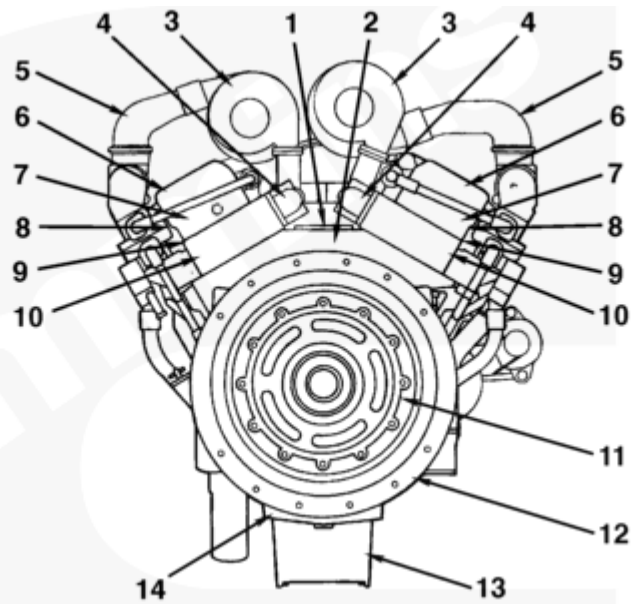
1. Turbochargers
2. Left-bank water outlet
3. Left-bank thermostat housing
4. Coolant filter head
5. Coolant filters
6. Fan pilot
7. Front gear cover
8. Accessory drive pulley
9. Direction of rotation
10. Crankshaft pulley
11. Vibration damper
12. Alternator drive pulley
13. Fan idler pulley
14. Fan hub
15. Right-bank thermostat housing
16. Right-bank water outlet.



ew600gn

Left View - KTA38 with Outboard Aftercoolers

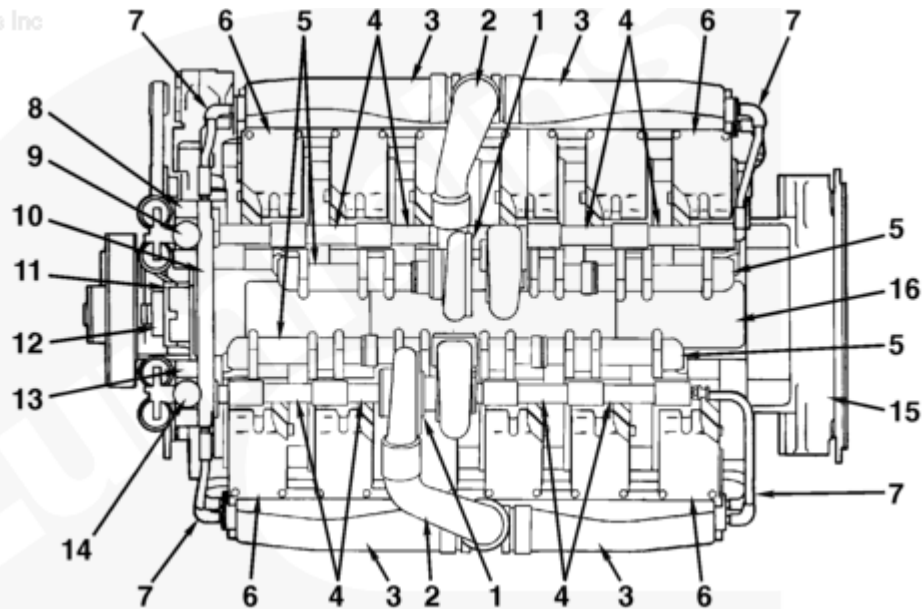
1. Turbochargers
2. Air connection
3. Air crossover connection
4. Aftercooler assembly
5. Rocker lever cover
6. Aftercooler water outlet tube
7. Left-bank water outlet
8. Thermostat housing support
9. Thermostat housing
10. Aftercooler water inlet tube
11. Cam follower cover
12. Front gear housing
13. Front gear cover
14. Air compressor drive
15. Air compressor
16. Dataplate
17. Fuel pump
18. Fuel filter head
19. Fuel filter
20. Full-flow lubricating oil filter
21. Crankcase breather
22. Oil level gauge
23. Lubricating oil filter head
24. Oil filler port
25. Flywheel housing.



00600379

Rear View - KTA38 with Outboard Aftercoolers

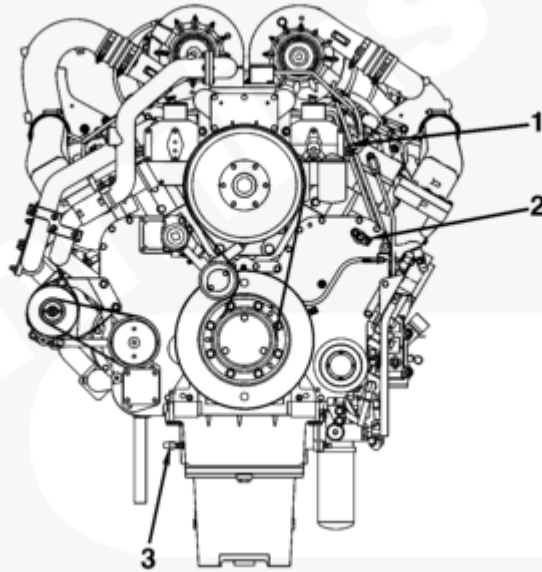
1. Oil cooler cover plate
2. Cylinder block
3. Turbocharger
4. Exhaust manifold
5. Air crossover connection
6. Rocker lever cover
7. Rocker lever housing
8. STC (HVT manifold)
9. Fuel manifold
10. Cylinder head
11. Flexplate
12. Flywheel housing
13. Oil pan
14. Oil pan adapter.



ew600go

Top View - KTA38 with Outboard Aftercoolers

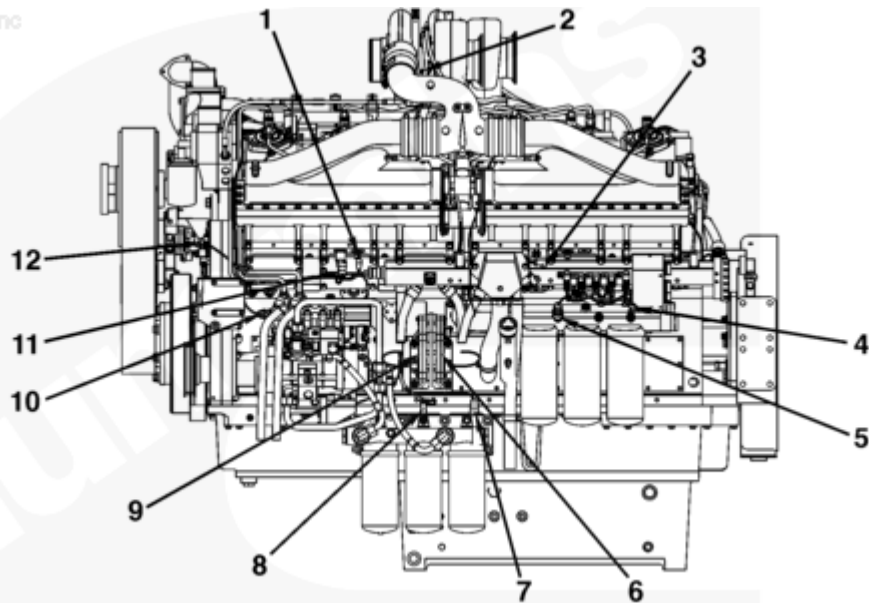
1. Turbocharger
2. Air crossover connection
3. Aftercooler assembly
4. Water tubes
5. Exhaust manifolds
6. Rocker lever cover
7. Aftercooler water outlet tube
8. Right-bank thermostat housing
9. Right-bank water outlet
10. Thermostat housing support
11. Fan hub support
12. Fan hub shaft
13. Left-bank thermostat housing
14. Left-bank water outlet
15. Flywheel housing
16. Oil cooler cover plate.



00600382

QSK38 Industrial - Front View Sensor Location

1. Engine coolant temperature sensor
2. Camshaft position sensor
3. Lubricating oil temperature sensor.



00600385

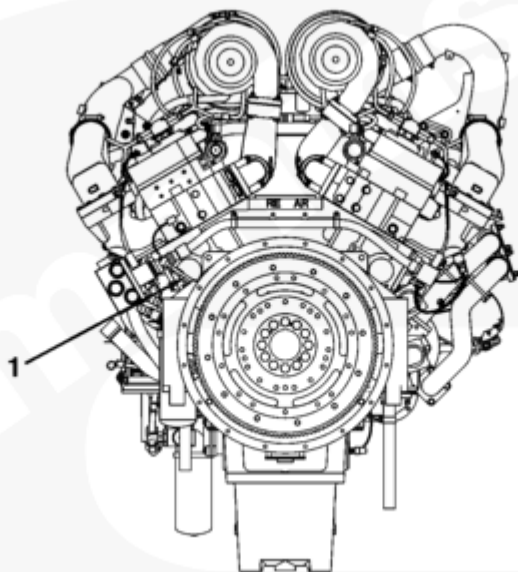
QSK38 Industrial - Left View Sensor Location

1. Intake manifold temperature sensor (left bank front)
2. Turbocharger speed sensor
3. Ambient air pressure sensor
4. Pre-oil rifle pressure sensor
5. Oil rifle pressure sensor
6. ECM 2 (source address 00)
7. Fuel supply temperature sensor
8. Fuel supply pressure sensor
9. ECM 1 (source address 01)

10. Fuel accumulator pressure sensor
11. Intake manifold pressure sensor
12. Camshaft position sensor

Note : The lubricating oil filter head can be left bank or right bank mounted.

©Cummins Inc

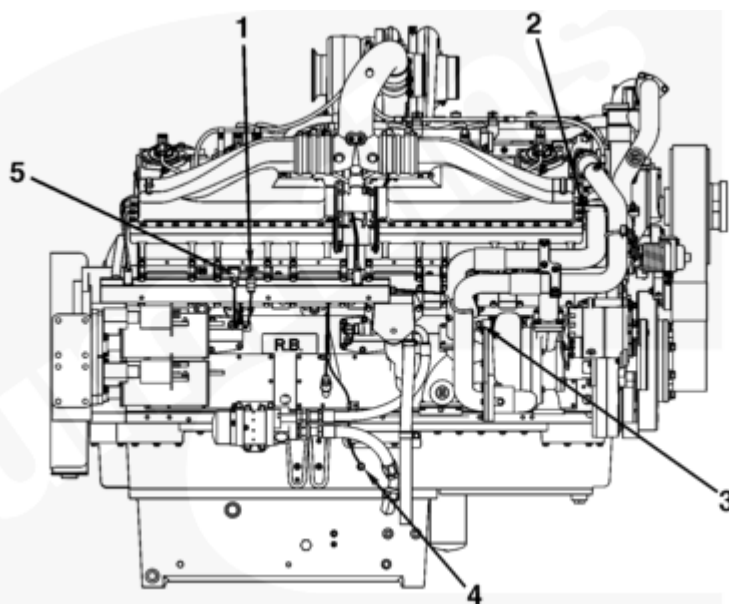


00600384

QSK38 Industrial - Rear View Sensor Location

1. Engine speed sensor (behind flywheel housing).

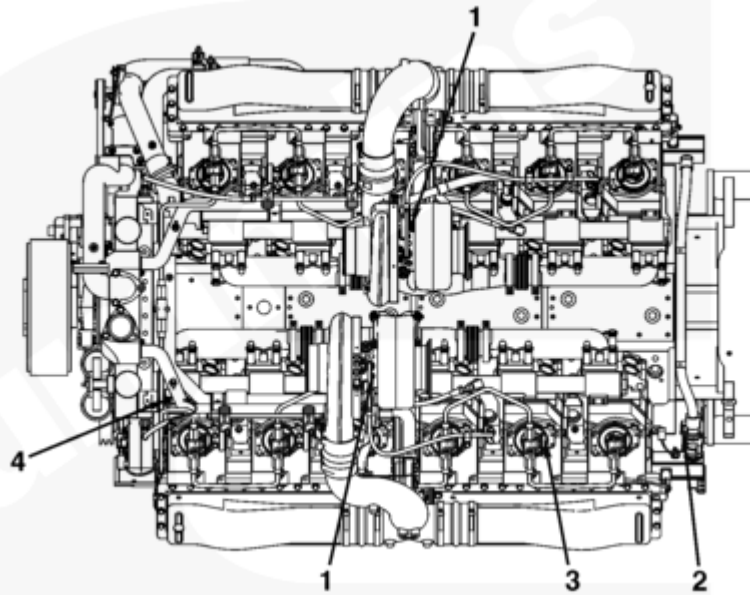
©Cummins Inc



00600383

QSK38 Industrial - Right View Sensor Location

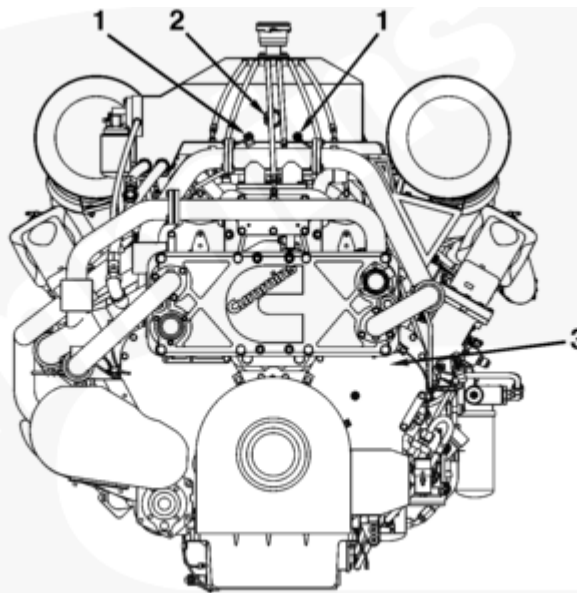
1. Intake manifold temperature sensor (right bank rear)
2. Crankcase pressure sensor
3. Coolant pressure sensor
4. Lubricating oil temperature sensor (lubricating oil pan)
5. Intake manifold pressure sensor.



00600386

QSK38 Industrial - Top View Sensor Location

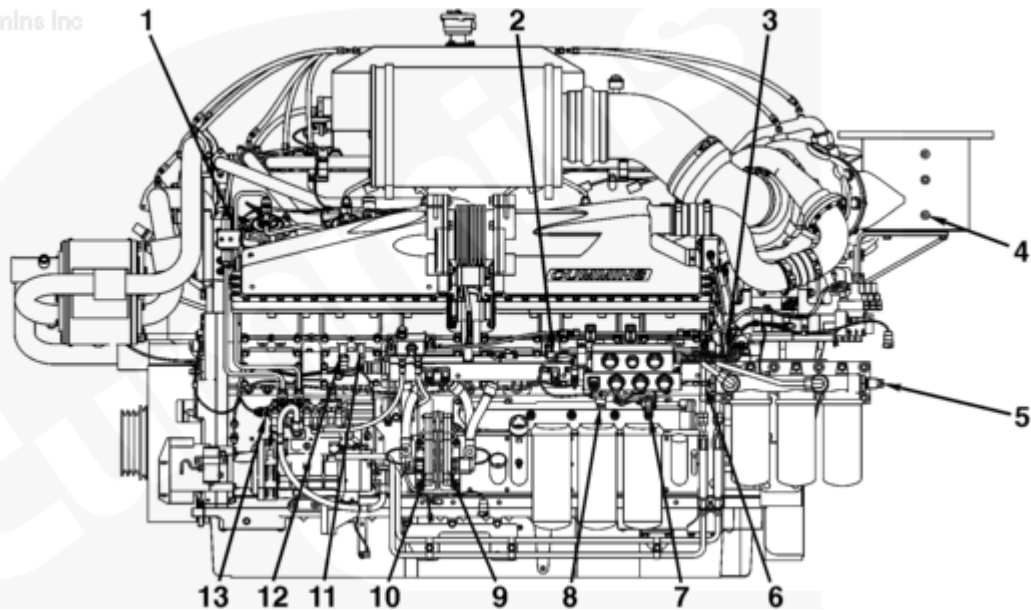
1. Turbocharger speed sensor
2. Engine speed sensor
3. Injector solenoid (one per cylinder on top of rocker cover)
4. Coolant temperature sensor.



00600391

QSK38 Marine with Heat Exchanger - Front View Sensor Location

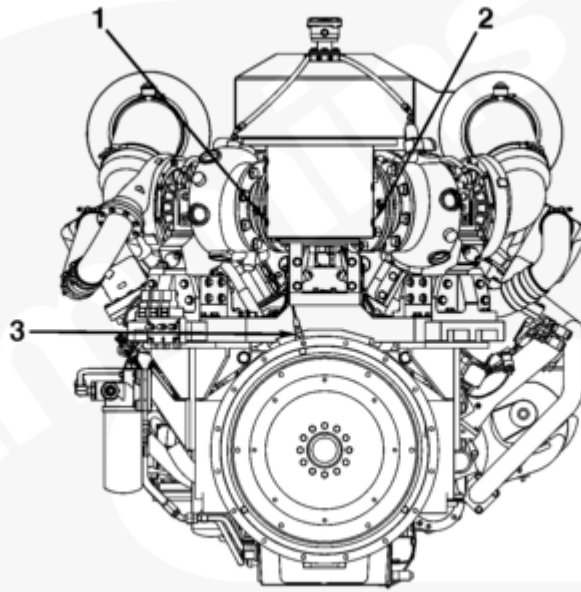
1. Coolant level sensor (top tank or OEM fitting)
2. Coolant sight glass
3. Camshaft position sensor.



00600394

QSK38 Marine with Heat Exchanger - Left View Sensor Location

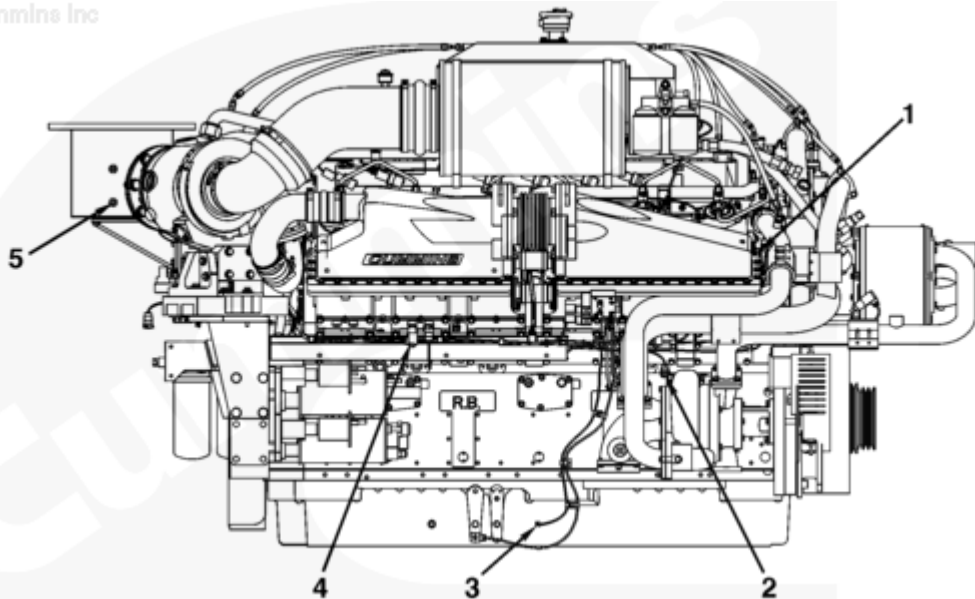
1. Engine coolant temperature sensor
2. Ambient air pressure sensor
3. Engine speed sensor
4. Engine exhaust gas stack temperature sensor, left manifold
5. Fuel supply pressure sensor
6. Fuel temperature sensor
7. Pre-oil filter pressure sensor
8. Oil rifle pressure sensor
9. ECM 2 (source address 00)
10. ECM 1 (source address 01)
11. Intake manifold temperature sensor (left bank front)
12. Intake manifold pressure sensor (left bank front)
13. Fuel accumulator pressure sensor.



00600393

QSK38 Marine with Heat Exchanger - Rear View Sensor Location

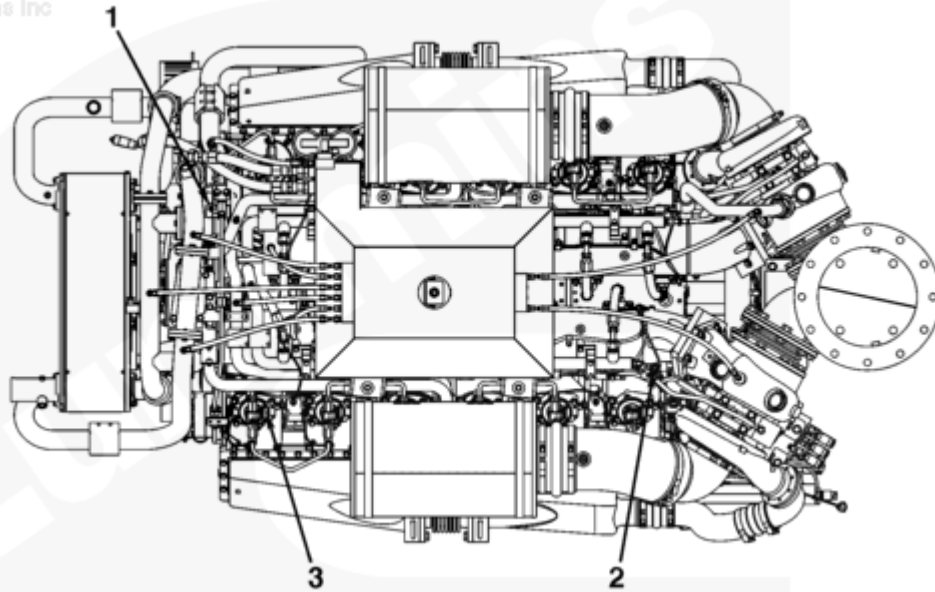
1. Engine exhaust gas stack temperature sensor, left manifold
2. Engine exhaust gas stack temperature sensor, right manifold
3. Engine speed sensor (behind the flywheel housing).



00600392

QSK38 Marine with Heat Exchanger - Right View Sensor Location

1. Crankcase pressure sensor
2. Coolant pressure sensor
3. Lubricating oil temperature sensor (oil pan)
4. Intake manifold pressure sensor (right bank rear)
5. Engine exhaust gas stack temperature sensor, right manifold.

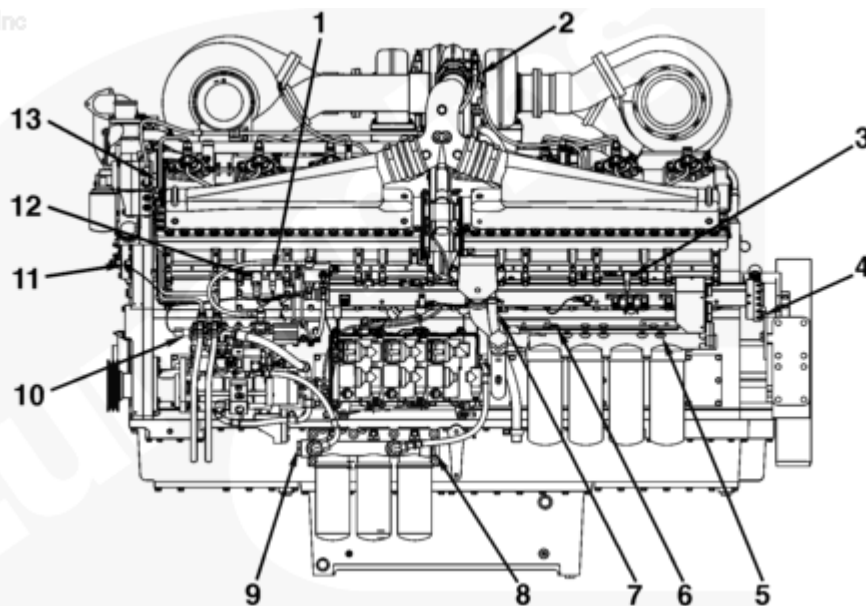


00600395

QSK38 Marine with Heat Exchanger - Top View Sensor Location

1. Crankcase pressure sensor (behind gear housing)
2. Exhaust gas temperature sensor (one per cylinder located in the exhaust port outlet of the cylinder head)
3. Injector solenoid (one per cylinder on top of the rocker cover).

Note : The exhaust gas temperature sensors on Marine engines are optional.



00600261

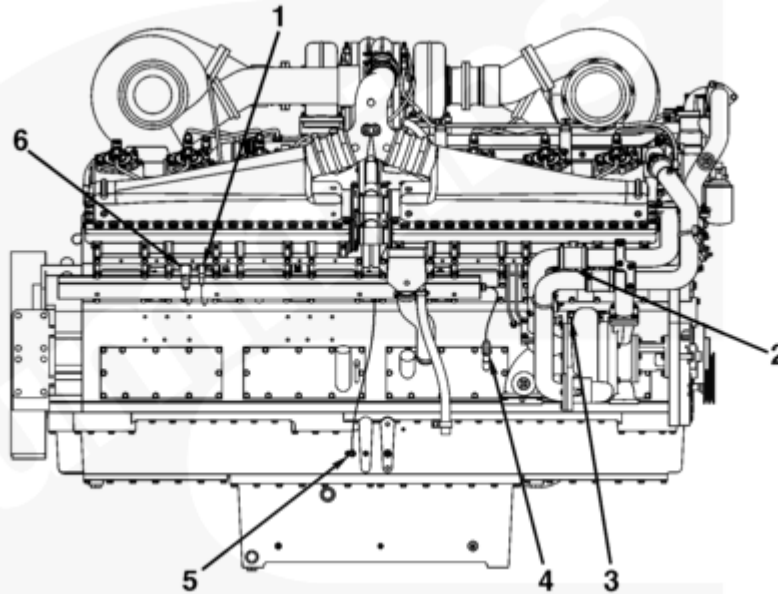
QSK50 Industrial - Left View Sensor Location

1. Intake manifold 1 temperature sensor
2. Turbocharger speed 1 sensor (for engines with single-stage **only**)
3. Intake manifold 3 temperature sensor
4. Engine crankshaft speed/position sensor
5. Engine oil rifle pressure 1 sensor (left-bank option)
6. Pre-oil filter pressure sensor
7. Barometric pressure sensor

8. Fuel delivery pressure sensor
9. Engine fuel temperature sensor 1
10. Injector metering rail 1 pressure sensor
11. Engine camshaft speed/position sensor
12. Intake manifold 1 pressure sensor
13. Engine coolant temperature 1 sensor.

Note : Exhaust gas temperature sensors (8 on left-bank, not shown)

©Cummins Inc

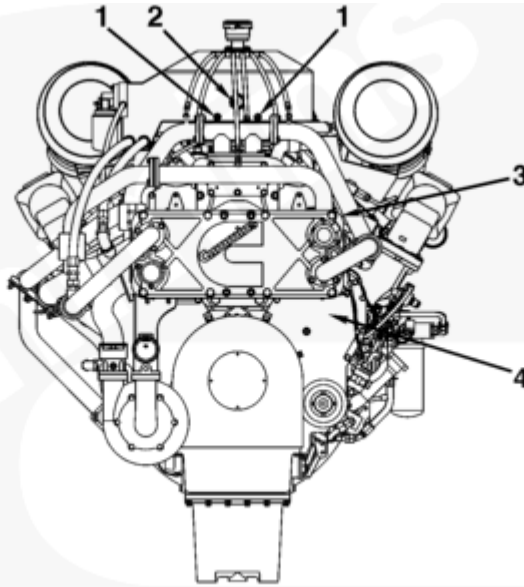


00600263

QSK50 Industrial - Right View Sensor Location

1. Intake manifold 4 temperature sensors
2. Intake manifold 2 temperature sensor
3. Coolant pressure sensor
4. Crankcase pressure sensor
5. Engine oil temperature sensor 1
6. Intake manifold 2 pressure sensor.

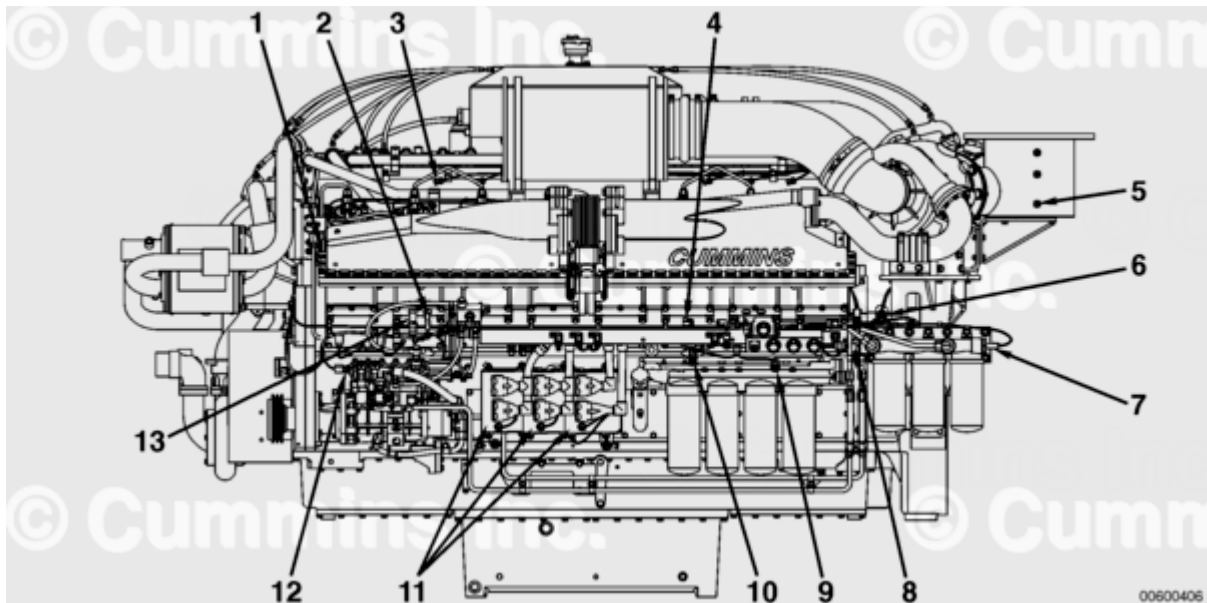
Note : Exhaust gas temperature sensors (8 on the right bank, not shown).



00600395

QSK50 Marine with Heat Exchanger - Front View Sensor Location

1. Engine coolant level sensors
2. Coolant sight glass
3. Engine coolant temperature sensor
4. Camshaft position sensor.



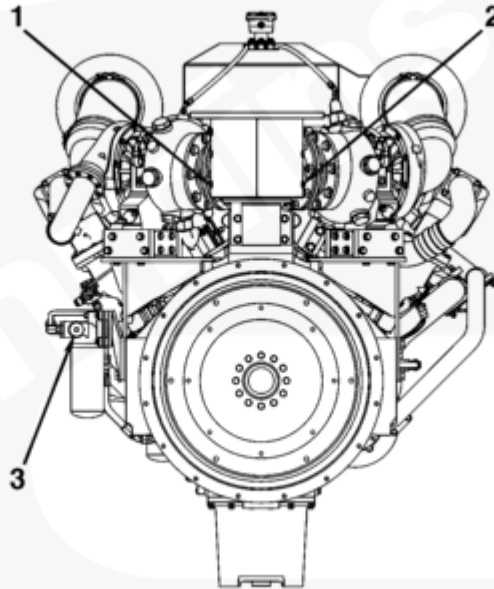
00600406

QSK50 Marine with Heat Exchanger- Left View Sensor Location

1. Coolant temperature sensor
2. Intake manifold temperature sensor, left bank front
3. Exhaust gas temperature sensor (one per cylinder located in the exhaust port outlet of the cylinder head)
4. Ambient air pressure sensor
5. Engine exhaust gas stack temperature sensor, left manifold
6. Engine speed sensor (behind flywheel housing)
7. Fuel supply pressure sensor

8. Fuel temperature sensor
9. Oil rifle pressure sensor
10. Pre-oil filter pressure sensor
11. ECMs (see last page of Engine Diagrams for identification)
12. Fuel accumulator pressure sensor
13. Intake manifold pressure sensor (left bank front).

©Cummins Inc

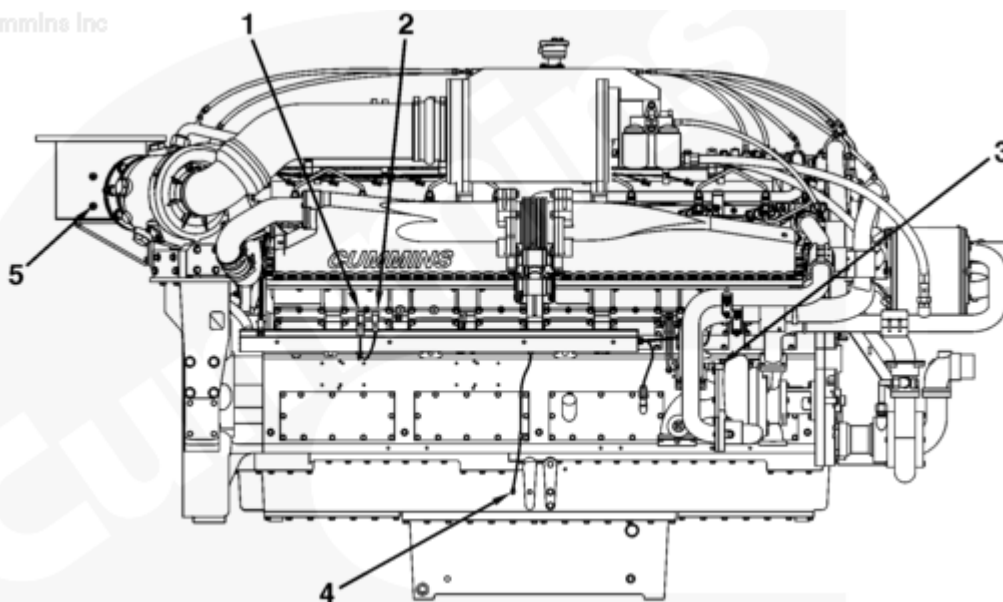


00600405

QSK50 Marine with Heat Exchanger - Rear View Sensor Location

1. Engine exhaust gas stack temperature sensor, left manifold
2. Engine exhaust gas stack temperature sensor, right manifold
3. Fuel supply pressure sensor.

©Cummins Inc



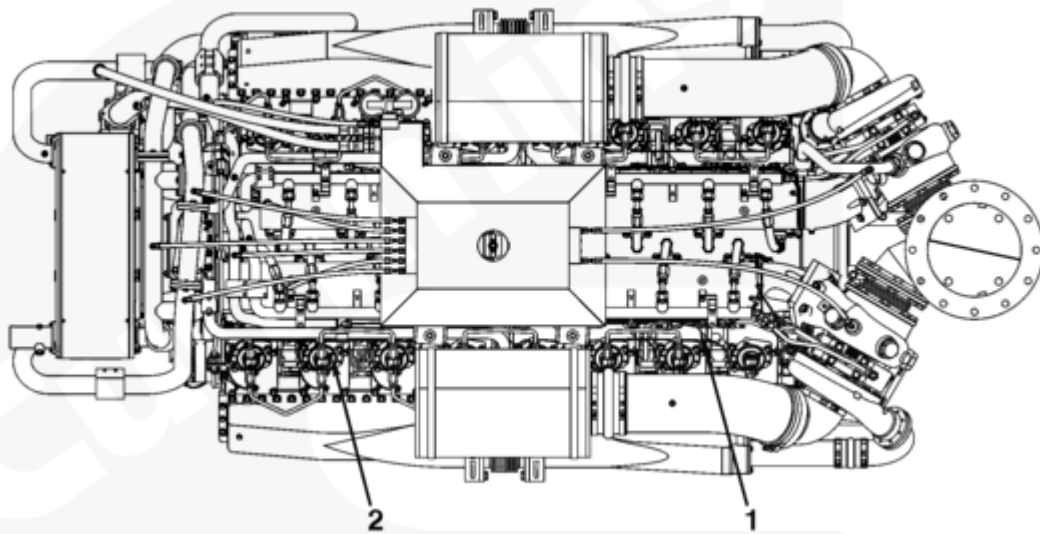
00600397

QSK50 Marine with Heat Exchanger - Right View Sensor Location

1. Intake manifold pressure sensor (rear right bank)
2. Intake manifold temperature sensor (rear right bank)

3. Engine coolant pressure sensor
4. Lubricating oil temperature sensor
5. Engine exhaust gas stack temperature sensor, right manifold.

©Cummins Inc



00600400

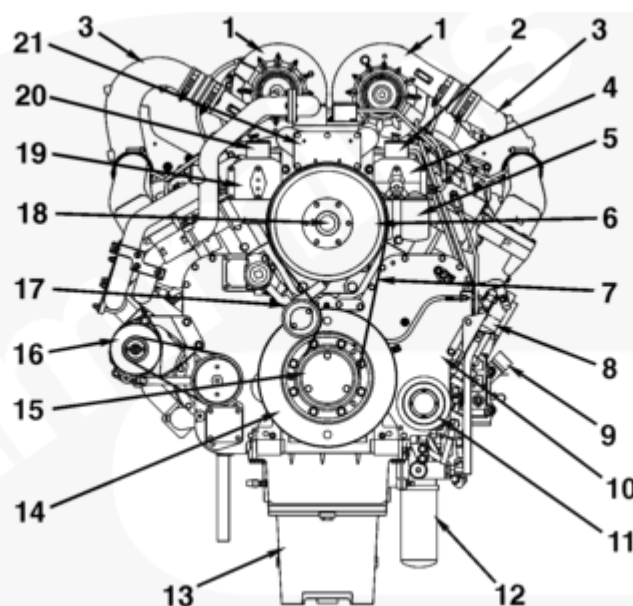
QSK50 Marine with Heat Exchanger- Top View Sensor Location

1. Exhaust gas temperature sensor (one per cylinder located in the exhaust port outlet of the cylinder head)
2. Injector solenoid (one per cylinder on top of the rocker housing cover).

Note : The exhaust gas temperature sensors on marine engines are optional sensors.

Note : These engine views are industrial with electronically actuated injectors.

©Cummins Inc

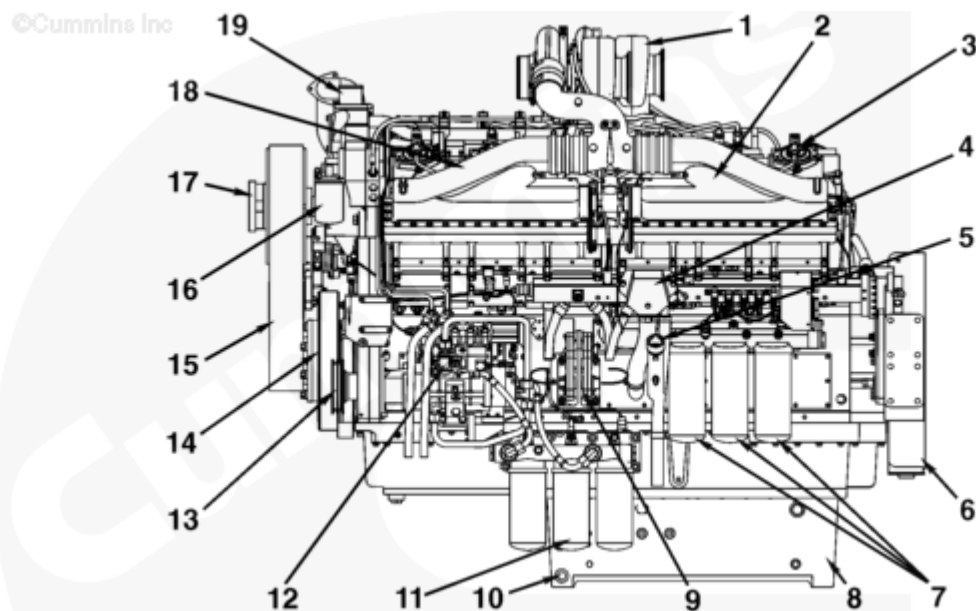


00600407

Front View - QSK38 Industrial (Electronically Actuated Injector)

1. Turbocharger
2. Left-bank coolant outlet

3. Air crossover pipe
4. Left-bank thermostat housing
5. Water filter
6. Fan pulley
7. Fan drive belt
8. Fuel pump
9. Lubricating oil fill
10. Front gear cover
11. Accessory drive pulley
12. Stage 2 fuel filters
13. Lubricating oil sump
14. Vibration damper
15. Camshaft pulley
16. Fan idler pulley
17. Alternator
18. Automatic belt tensioner
19. Fan hub
20. Right-bank coolant outlet
21. Right-bank thermostat housing



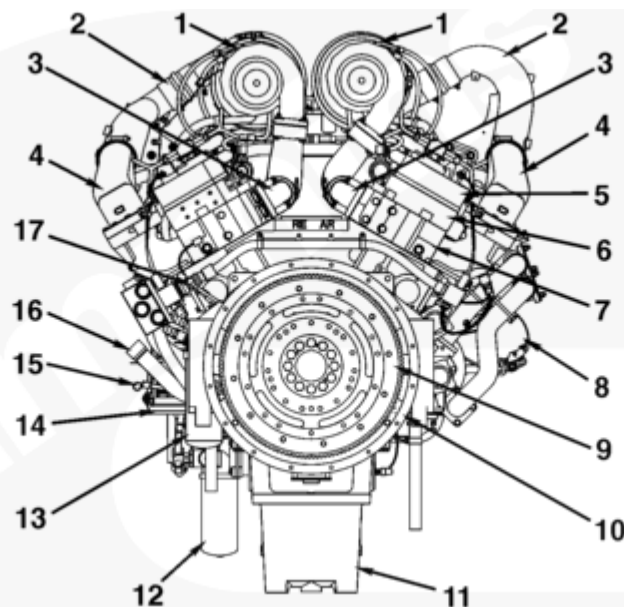
00600410

Left View - QSK38 Industrial (Electronically Actuated Injector)

1. Turbocharger
2. Aftercooler
3. Injector
4. Crankshaft breather
5. Lubricating oil fill
6. Flywheel housing
7. Lubricating oil filters
8. Lubricating oil sump
9. ECMs

10. Lubricating oil sump drain
11. Stage 2 fuel filters
12. Fuel pump
13. Accessory drive pulley
14. Vibration damper
15. Fan drive belt
16. Water filter
17. Fan hub
18. Air crossover pipe
19. LTA coolant outlet.

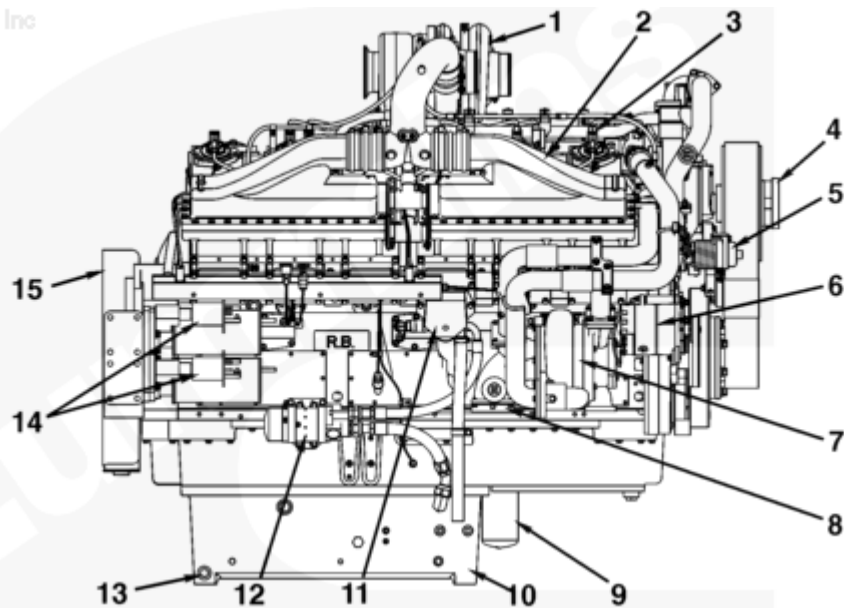
©Cummins Inc



00600409

Rear View - QSK38 Industrial (Electronically Actuated Injector)

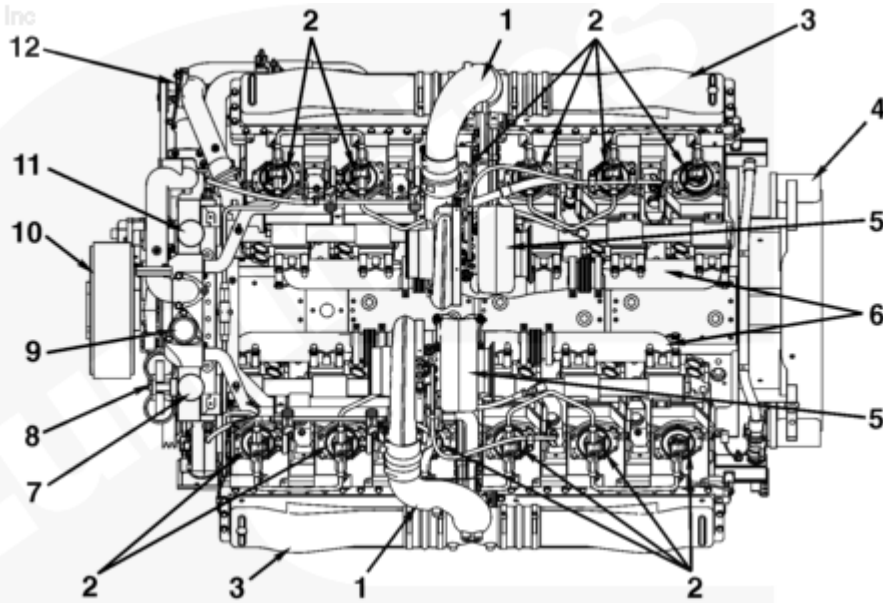
1. Turbochargers
2. Air crossover pipe
3. Exhaust manifolds
4. Aftercooler covers
5. Rocker lever cover
6. Rocker lever housing
7. Cylinder head
8. Alternator
9. Flexplate
10. Flywheel housing
11. Oil sump
12. Stage 2 fuel filter
13. Lubricating oil filter
14. ECMs
15. Dipstick
16. Lubricating oil fill
17. Engine speed sensor.



00600408

Right View - QSK38 Industrial (Electronically Actuated Injector)

1. Turbocharger
2. Aftercooler cover
3. Injector
4. Fan hub
5. Fan belt tensioner
6. Alternator
7. Water pump
8. Water pump coolant inlet
9. Stage 2 fuel filter
10. Lubricating oil sump
11. Crankcase breather
12. Pre-lubrication oil pump
13. Lubricating oil sump drain
14. Starters
15. Flywheel housing.

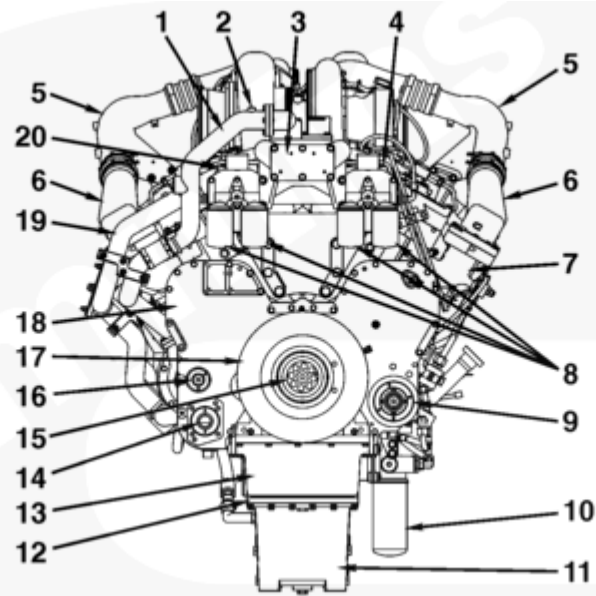


00600411

Top View - QSK38 Industrial (Electronically Actuated Injectors)

1. Air crossovers
2. Injectors
3. Aftercooler covers
4. Flywheel housing
5. Turbochargers
6. Exhaust manifolds
7. Left-bank coolant outlet
8. Coolant filters
9. LTA Coolant outlet
10. Fan hub
11. Right-bank coolant outlet
12. Alternator.

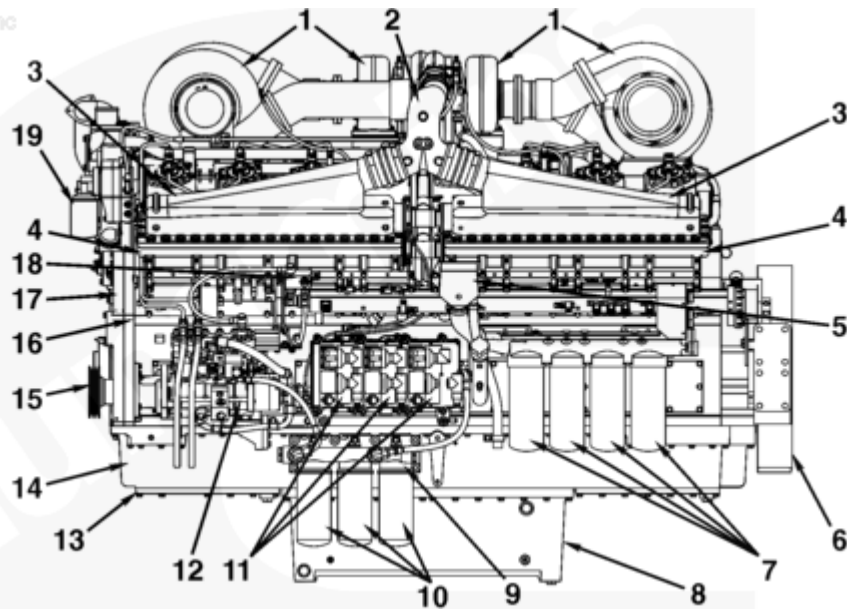
with Electronically Actuated Injector



05600290

Front View - QSK50

1. Low temperature aftercooler thermostat bypass tube
2. Low temperature aftercooler vent tube
3. Low temperature aftercooler thermostat housing
4. High temperature vent line
5. Air crossover connections
6. Aftercooler cover
7. Intake manifold
8. Coolant filters
9. Accessory drive pulley
10. Stage 2 fuel filter
11. Oil pan sump
12. Oil pan cover plate
13. Oil pan adapter
14. Accessory drive
15. Crankshaft pulley
16. Alternator drive pulley
17. Vibration dampers
18. Front gear cover
19. Water pump outlet tube
20. High temperature vent line.

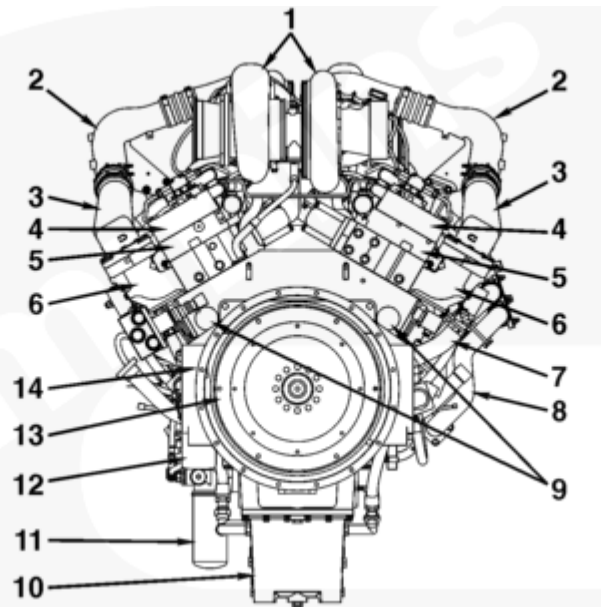


00600287

Left View - QSK50

1. Turbochargers
2. Air crossover connections
3. Aftercooler covers
4. Intake manifold
5. Crankcase breather
6. Flywheel housing
7. Lubricating oil filters
8. Oil pan sump
9. Fuel filter head
10. Stage 2 fuel filters
11. Electronic Control Modules (ECM)
12. Fuel pump
13. Oil pan cover plate
14. Oil pan adapter
15. Accessory drive pulley
16. Front gear housing
17. Front gear cover
18. Fuel drain manifold block
19. Coolant filter.

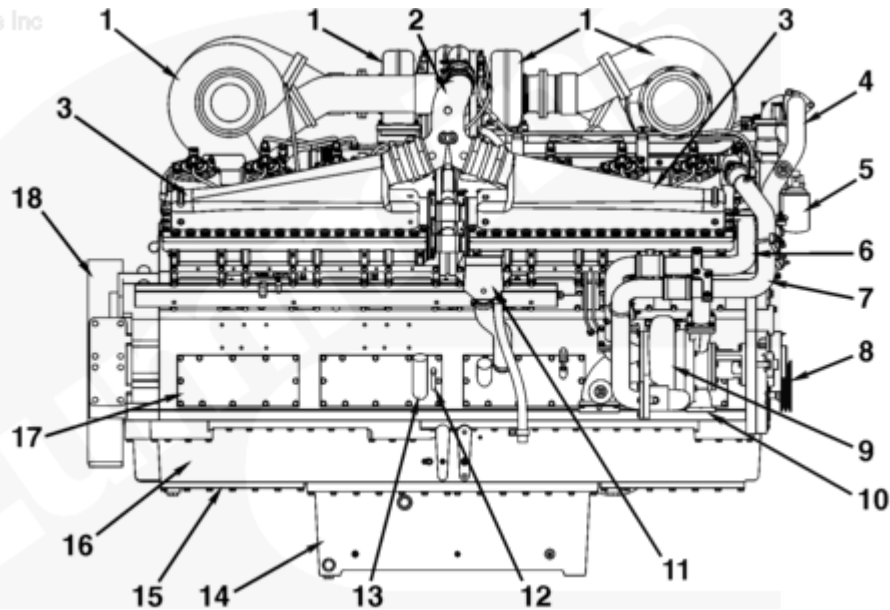
Note : Fuel filter (1st stage) is located off engine and is mounted by the OEM.



05600289

Rear View - QSK50

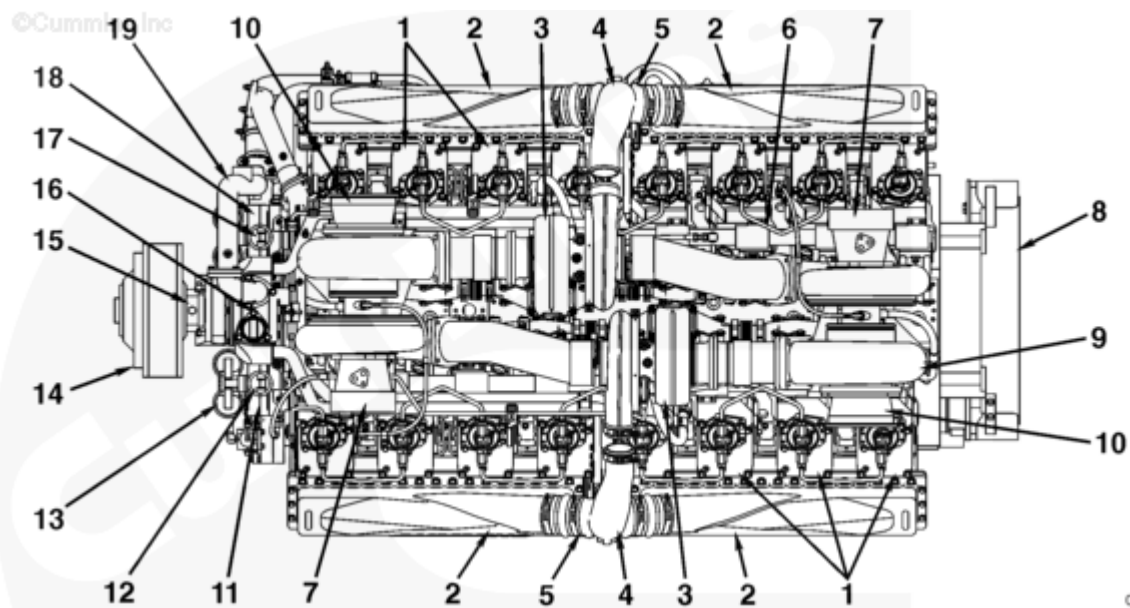
1. Turbochargers
2. Air crossover connections
3. Aftercooler covers
4. Rocker lever covers
5. Rocker lever housing
6. Intake manifold
7. Jacket water bypass tube
8. Water pump outlet tube
9. Camshaft covers
10. Oil pan sump
11. Stage 2 fuel filter
12. Lubricating oil filters
13. Flywheel
14. Flywheel housing.



00600257

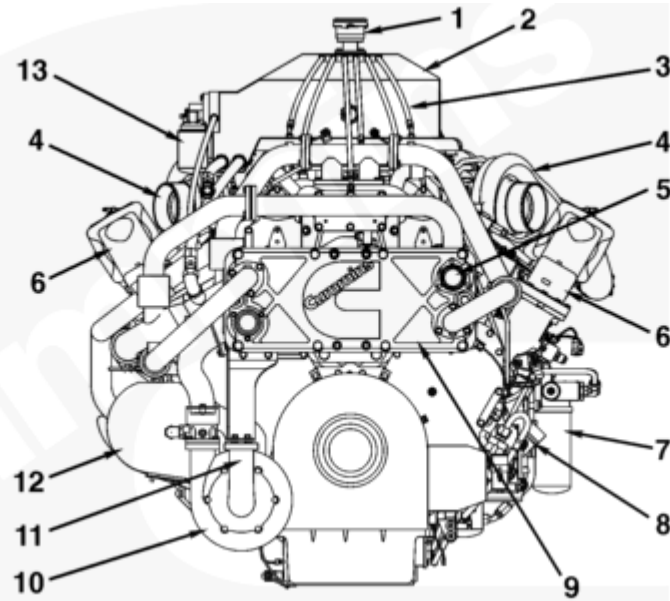
Right View - QSK50

1. Turbochargers
2. Air crossover connections
3. Aftercooler covers
4. Low temperature aftercooler thermostat bypass tube
5. Coolant filter
6. Jacket water bypass tube
7. Water pump outlet tube
8. Accessory drive pulley
9. Water pump
10. Low temperature aftercooler water pump supply inlet
11. Crankcase breather
12. Dipstick tube
13. Oil fill
14. Oil pan sump
15. Oil pan cover plate
16. Oil pan adapter
17. Hand hole covers
18. Flywheel housing.



Top View - QSK50

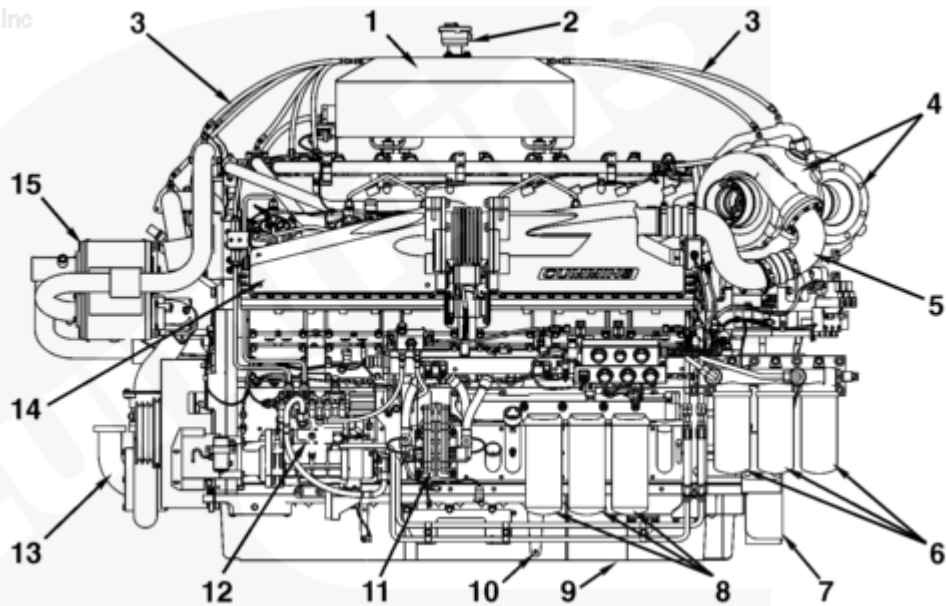
1. Rocker lever cover
2. Aftercooler cover
3. High stage turbocharger
4. Air crossover connection
5. Air crossover
6. Water transfer tube
7. Air inlet to turbocharger
8. Flywheel housing
9. Low stage turbocharger
10. Exhaust outlet
11. Left-bank thermostat housing
12. Left-bank water outlet
13. Coolant filters (2 on each thermostat housing)
14. Fan hub
15. Fan hub support
16. Low temperature aftercooler thermostat housing
17. Right-bank water outlet
18. Right-bank thermostat housing
19. Low temperature aftercooler tube.



00600335

Front View - QSK38 Marine with Heat Exchanger

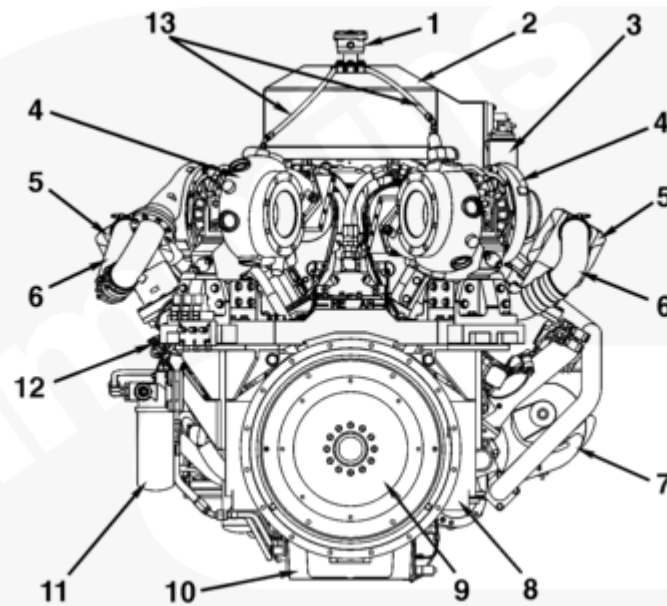
1. Coolant fill
2. Expansion tank
3. Coolant vent lines
4. Turbocharger
5. Sea water outlet
6. Aftercooler covers
7. Stage 2 fuel filter
8. Fuel pump
9. Heat exchanger
10. Sea water pump
11. Sea water inlet
12. Alternator belt guard
13. Coolant filters.



00600338

Left View - QSK38 Marine with Heat Exchanger

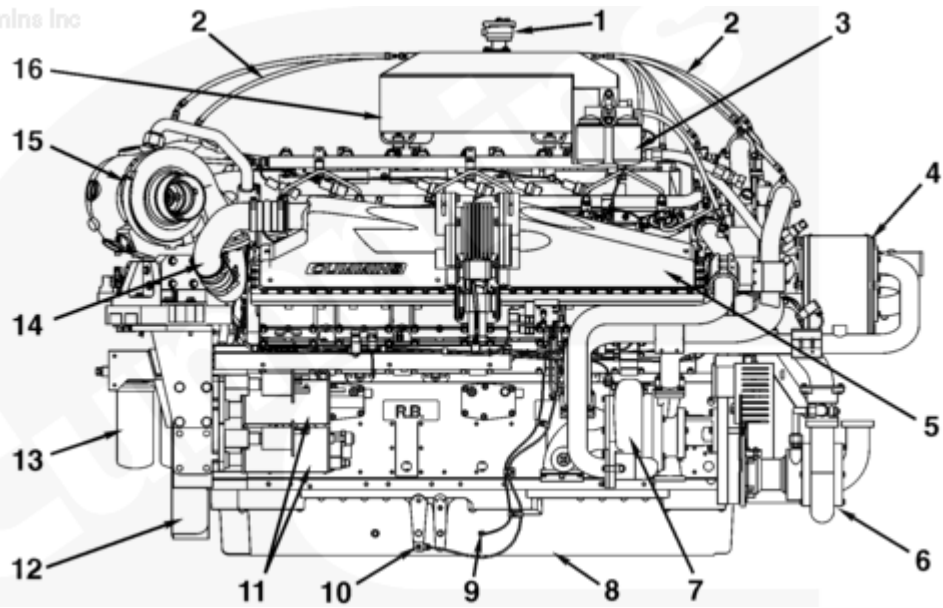
1. Expansion tank
2. Coolant fill
3. Coolant vent lines
4. Turbochargers
5. Air crossover pipe
6. Stage 2 fuel filters
7. Flywheel housing
8. Lubrication oil filters
9. Oil sump
10. Oil sump drain
11. ECMs
12. Fuel pump
13. Sea water pump
14. Aftercooler cover
15. Heat exchanger.



00600337

Rear View - QSK38 Marine with Heat Exchanger

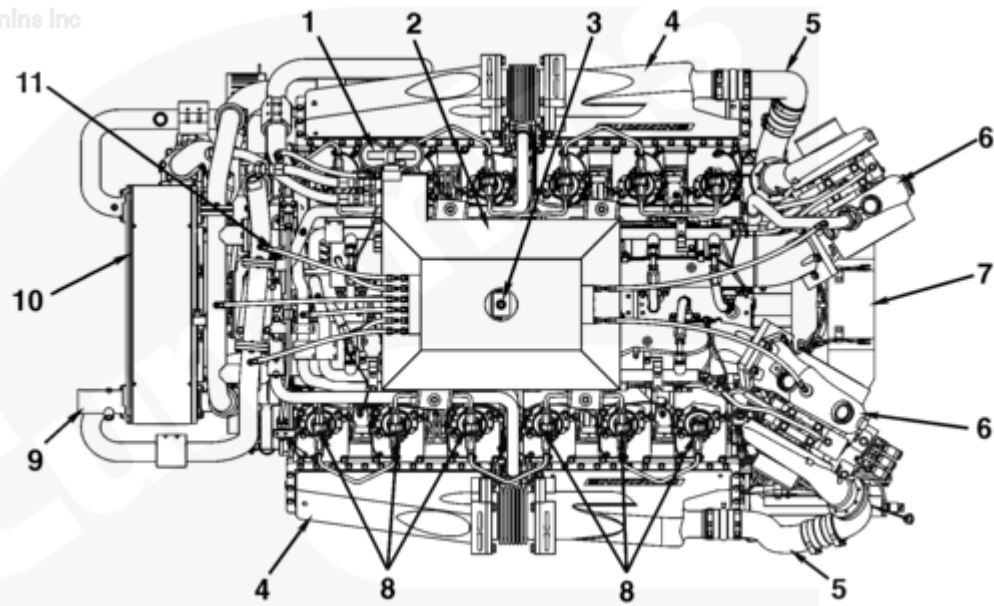
1. Coolant fill
2. Expansion tank
3. Coolant filters
4. Turbochargers
5. Aftercooler covers
6. Air crossover pipe
7. Alternator
8. Flywheel housing
9. Flywheel
10. Oil sump
11. Stage 2 fuel filters
12. Fuel pump
13. Coolant vent lines.



00600336

Right View - QSK38 Marine with Heat Exchanger

1. Coolant fill
2. Coolant vent lines
3. Coolant filters
4. Heat exchanger
5. Aftercooler cover
6. Sea water pump
7. Engine water pump
8. Oil sump
9. Oil level sensor
10. Oil heater
11. Starters
12. Flywheel housing
13. Stage 2 fuel filters
14. Air crossover pipe
15. Turbocharger
16. Expansion tank.

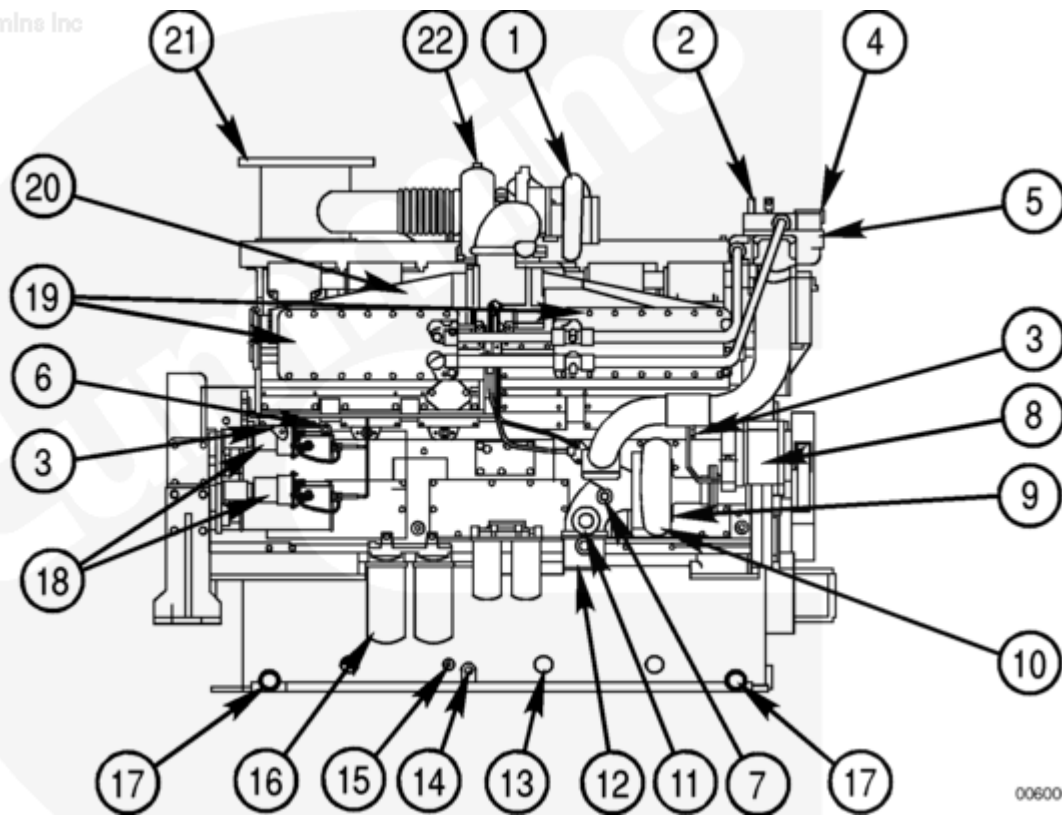


00600339

Top View - QSK38 Marine with Heat Exchanger

1. Coolant filters
2. Expansion tank
3. Coolant fill
4. Aftercooler covers
5. Air crossover
6. Turbochargers
7. Flywheel housing
8. Injectors
9. Sea water outlet from heat exchanger
10. Heat exchanger
11. Coolant vent lines.

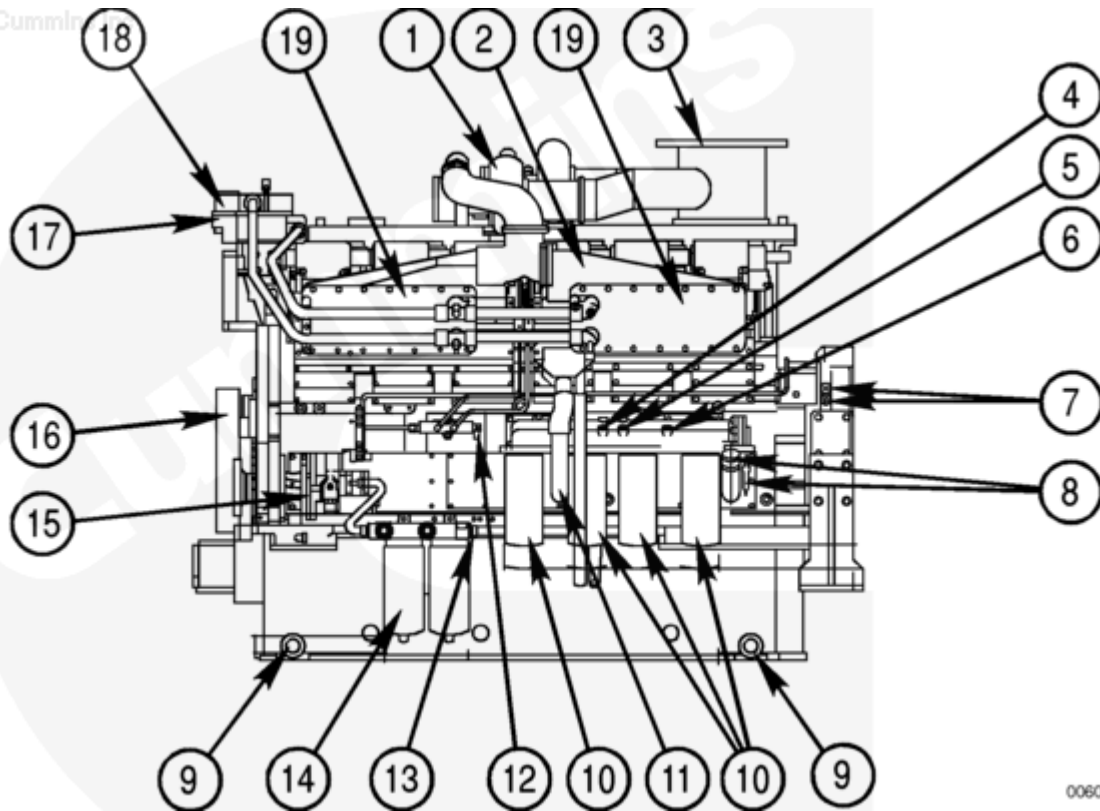
Marine Applications



00600004

Right View

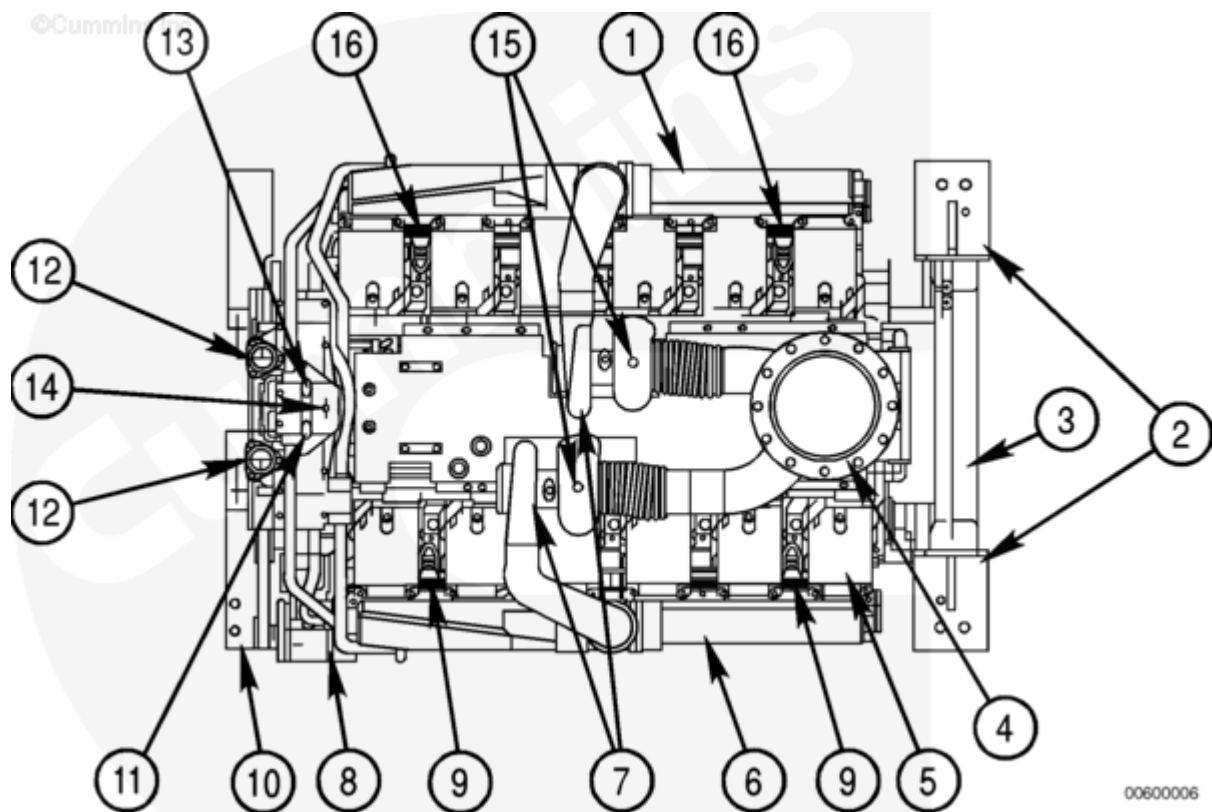
1. Turbocharger
2. Coolant vent line connections
3. Cylinder block drains
4. Dual water outlet connections
5. Thermostat housing
6. Coolant heater return
7. Coolant suction
8. Alternator
9. Water pump
10. Water pump drain
11. Coolant heater supply
12. Coolant inlet connection
13. Oil immersion heater
14. Oil temperature transducer
15. Bypass oil filter return
16. Bypass oil filter
17. Engine oil drains
18. Starters
19. Aftercoolers
20. Intake manifold
21. Exhaust
22. Turbocharger coolant vent line connections.



00600005

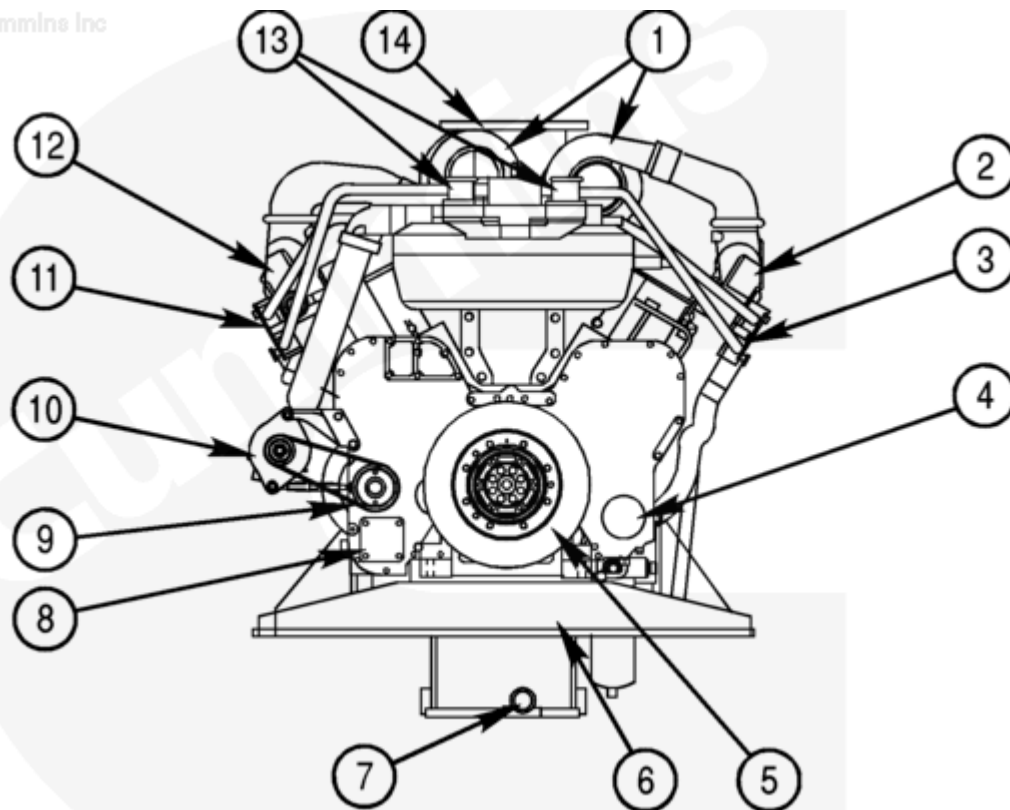
Left View

1. Turbocharger
2. Intake manifold
3. Exhaust
4. Fuel return to tank
5. Fuel inlet connection
6. Monitoring switch (filtered oil) pre-low oil pressure
7. Monitoring switch (filtered oil) low oil pressure
8. Oil pressure transducer
9. Magnetic pickups
10. Oil fill and dipstick
11. Oil filters
12. Engine oil drains
13. Crankcase breather
14. Fuel filters
15. Fuel pump
16. Vibration damper
17. Thermostat housing
18. Dual water outlet connections
19. Aftercoolers.



Top View

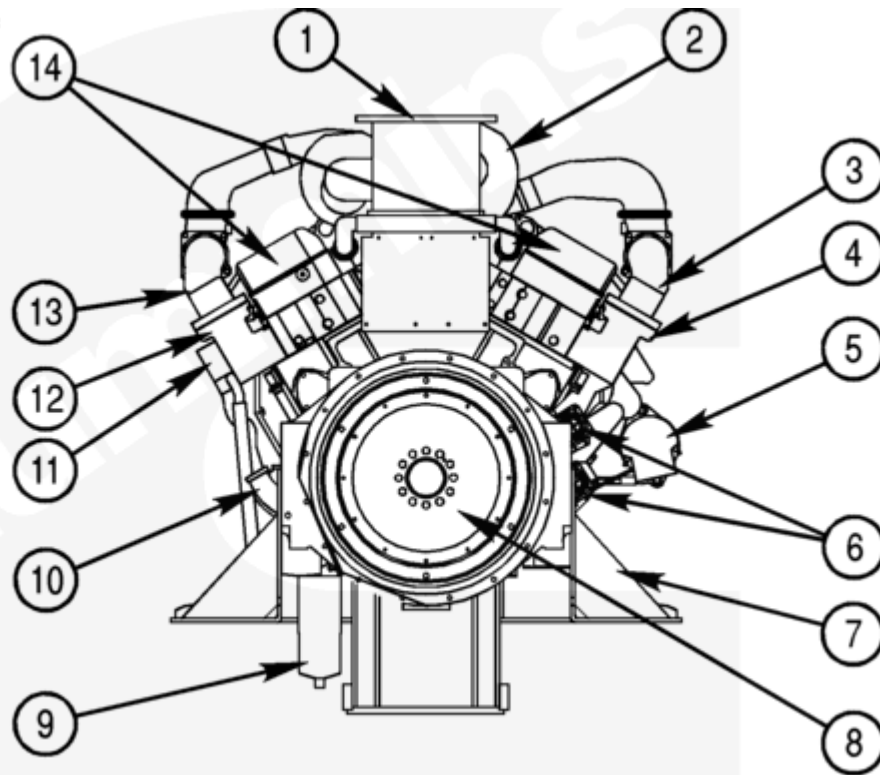
1. Right-bank intake manifold
2. Rear engine mounts
3. Flywheel housing
4. Exhaust
5. Rocker lever cover
6. Left-bank intake manifold
7. Turbochargers
8. Alternator
9. Lifting brackets
10. Front engine mount
11. Aftercooler coolant vent connection
12. Dual water outlet connection
13. Engine coolant vent line connection
14. Coolant vent line connection-non-activated Low Temperature Aftercooler cavity
15. Turbocharger coolant vent line connection
16. Lifting brackets.



00600007

Front View

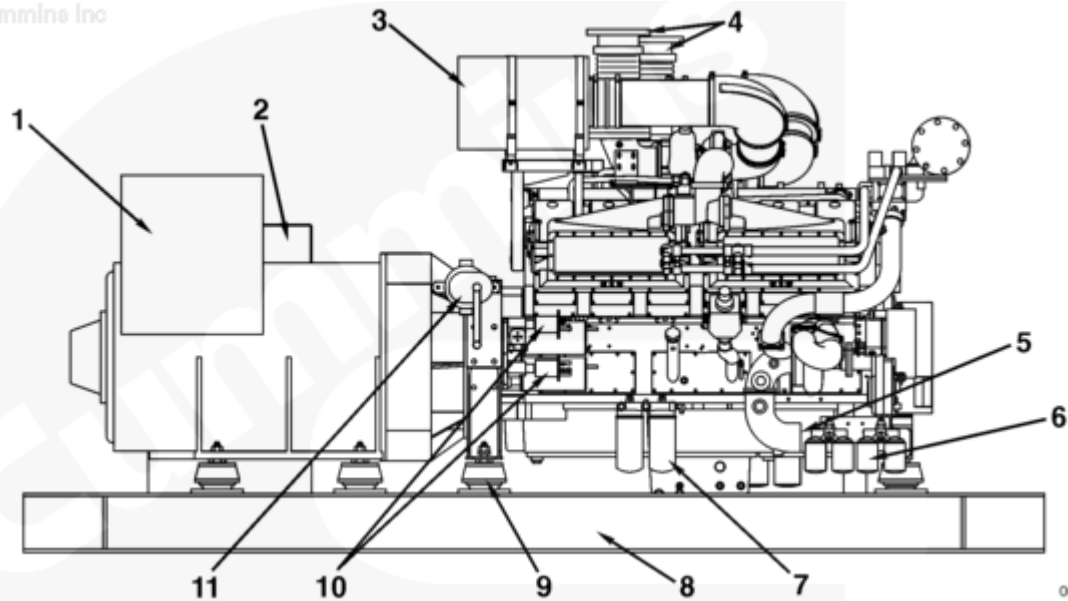
1. Turbochargers
2. Left intake manifold
3. Left side aftercooler
4. Pulley, accessory drive
5. Vibration damper
6. Engine front mount
7. Oil drain plug
8. Port, hydraulic pump drive
9. Alternator
10. Alternator drive
11. Right side aftercooler
12. Right intake manifold
13. Coolant outlet
14. Exhaust.



00600008

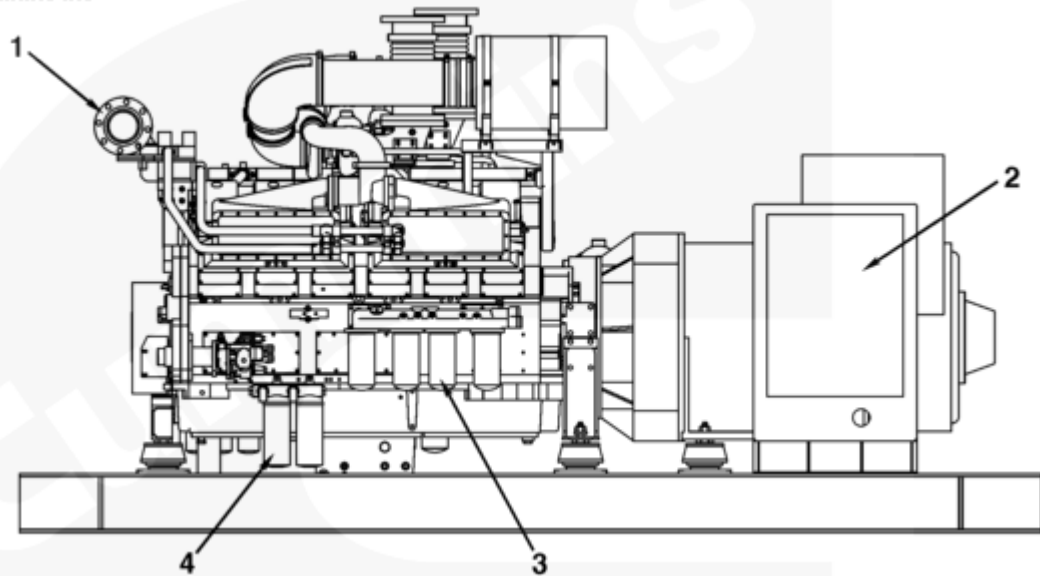
Rear View

1. Turbochargers
2. Left intake manifold
3. Left side aftercooler
4. Pulley, accessory drive
5. Vibration damper
6. Engine front mount
7. Oil drain plug
8. Port, hydraulic pump drive
9. Alternator
10. Alternator drive
11. Right side aftercooler
12. Right intake manifold
13. Coolant Outlet
14. Exhaust.



Right View - K38 C Power Keel Cooler Cooled (K50 Similar)

1. Alternator
2. Local control panel
3. Air cleaners
4. Exhaust connections
5. Coolant inlet connection from keel cooler
6. Coolant filters
7. Duplex fuel filters
8. Base rail
9. Flexible mount
10. Starters
11. Lubricating oil sump hand pump.

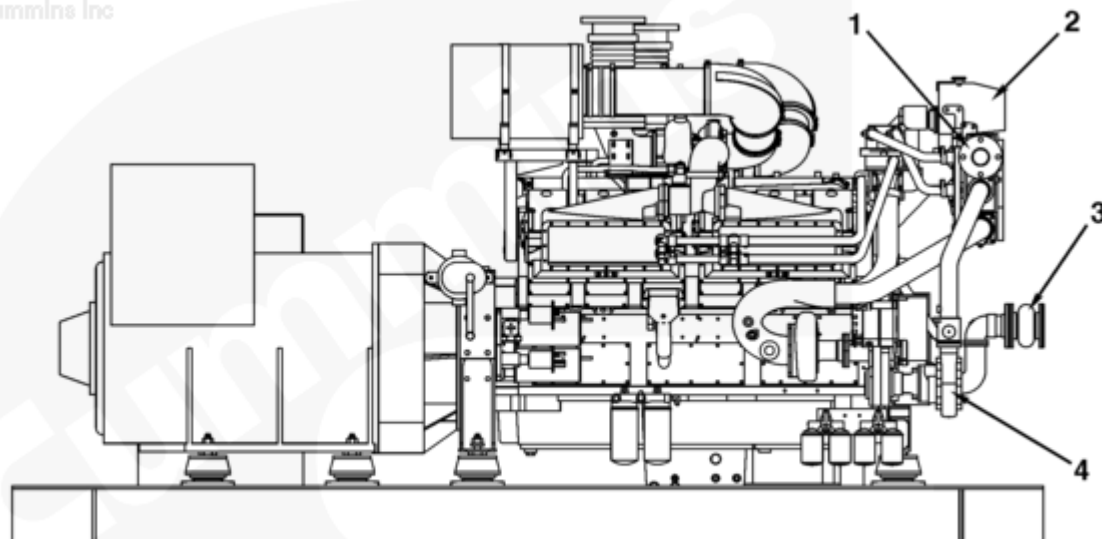


Left View - K38 C Power Keel Cooler Cooled

1. Coolant outlet connection to keel cooler

2. Local control panel
3. Duplex lubricating oil filters
4. Duplex fuel filters.

©Cummins Inc

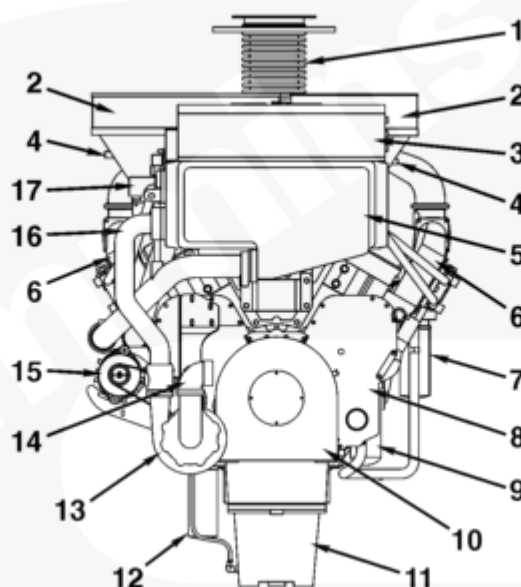


00600312

Right View - K38 C Power Heat Exchanger Cooled (K50 Similar)

1. Sea water outlet connection
2. Heat exchanger
3. Sea water inlet connection
4. Sea water pump.

©Cummins Inc

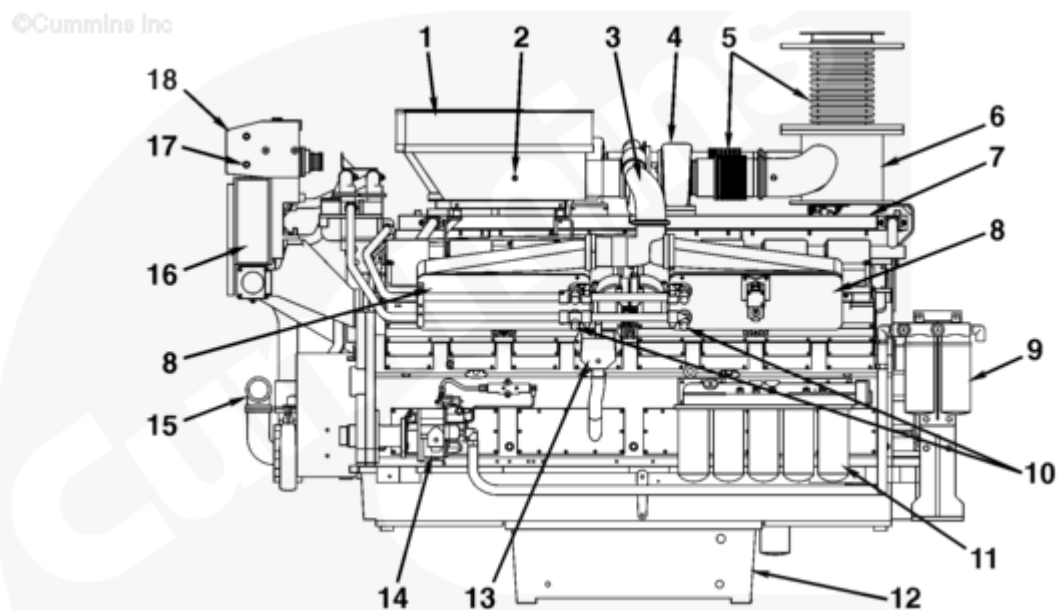


00600290

Front View - K50M2

1. Exhaust bellows
2. Air filter
3. Expansion tank
4. Air restriction indicator

5. Heat exchanger
6. Aftercooler
7. Fuel filters
8. Gear cover
9. Full-flow lubricating oil filters
10. Damper guard
11. Oil pan
12. Bypass lubricating oil filters
13. Sea water pump
14. Sea water inlet pipe
15. Alternator
16. Heat exchanger sea water inlet pipe
17. Sea water outlet connection.

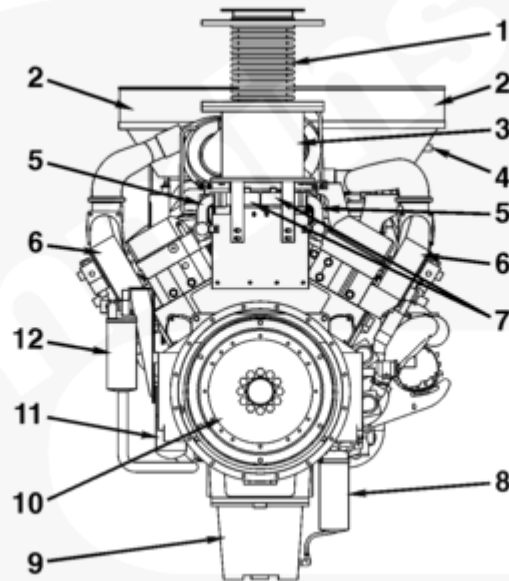


Left View - K50M2

1. Air filters
2. Air restriction indicator
3. Air crossover connection
4. Turbocharger
5. Exhaust bellows
6. Single exhaust collector
7. Heat shield
8. Aftercooler
9. Fuel filters
10. Aftercooler core drain
11. Full-flow lubricating oil filters
12. Oil pan
13. Crankcase breather
14. Fuel pump
15. Sea water inlet pipe

- 16. Heat exchanger
- 17. Sight glass
- 18. Expansion tank.

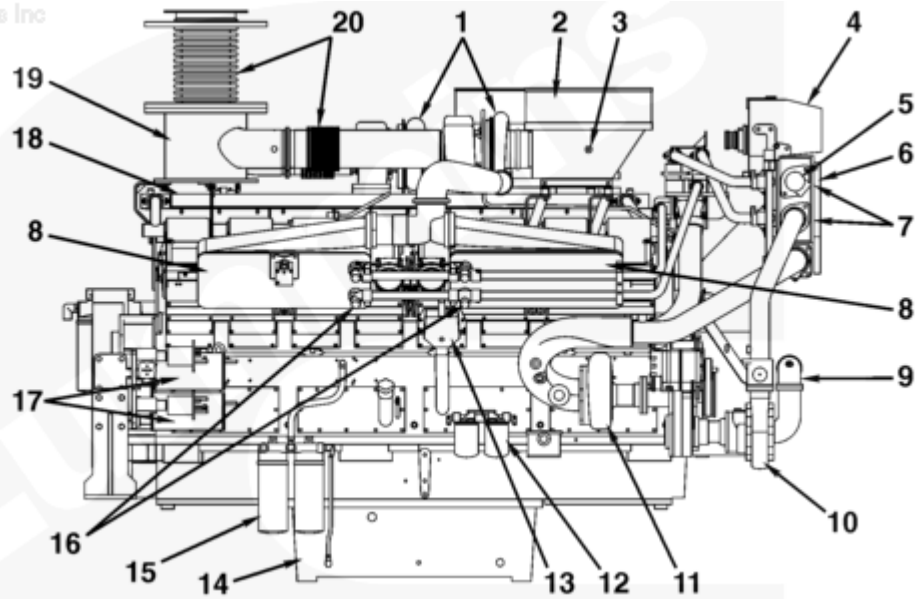
©Cummins Inc



00600231

Rear View - K50M2

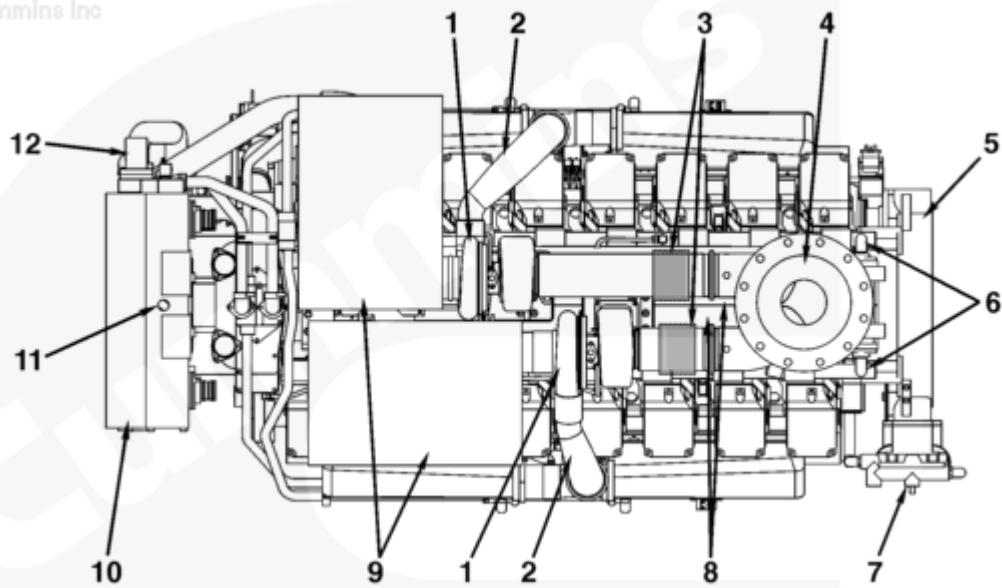
- 1. Exhaust bellows
- 2. Air filter
- 3. Single exhaust collector
- 4. Air restriction indicator
- 5. Heat shield coolant supply tube
- 6. Aftercooler
- 7. Heat shields
- 8. Bypass lubricating oil filters
- 9. Oil pan
- 10. Flywheel
- 11. Full-flow lubricating oil filters
- 12. Fuel filter.



00600227

Right View - K50M2

1. Turbocharger
2. Air filters
3. Air restriction indicator
4. Expansion tank
5. Sea water outlet connection
6. Heat exchanger
7. Zinc anode plugs
8. Aftercooler
9. Sea water inlet pipe
10. Sea water pump
11. Water pump
12. Coolant filters
13. Crankcase breather
14. Oil pan
15. Bypass lubricating oil filters
16. Aftercooler core drain
17. Starter
18. Heat shield
19. Single exhaust collector
20. Exhaust bellows.



00600229

Top View - K50M2

1. Turbocharger
2. Air crossover connection
3. Exhaust bellows
4. Single exhaust collector
5. Flywheel housing
6. Heat shield coolant supply tubes
7. Fuel filter head
8. Heat shield
9. Air filters
10. Expansion tank
11. Pressure cap
12. Sea water outlet connection.